Hypatia Library
Clean Articles

Hypatia exists at the intersection of AI and psychological research. The present document contains articles in both these fields. Citations, parenthetical references to figures/tables, headers, footers, and dense statistical data (like p-values and standard deviations) have been removed to ease the reading process. The complete articles can be found in the "Hypatia: Articles" folder.

ARTICLES

Article 1: Assessing personality using zero-shot generative AI scoring of brief open-ended text (Wright 2026)

Abstract: Researchers show that widely available large language models (LLMs) can — out of the box — accurately score people's personality traits on the basis of their brief, open-ended narratives. LLM ratings converged with self-reports, predicted daily behavior and mental health, and outperformed traditional language processing methods. Thus, use of LLM tools emerges as an accurate, scalable and efficient approach to assessing arbitrary psychological constructs. This study evaluates the capability of commercially available Large Language Models (LLMs), such as ChatGPT and Claude, to assess Big-Five personality traits directly from open-ended narratives without specific model training (zero-shot learning). Using two distinct samples—one involving spontaneous streams of thought and another involving daily video diaries—the authors prompted seven different LLMs to complete standard personality inventories based on the participants' transcripts. The results demonstrated that LLM-generated trait scores achieved high convergence with participants' self-reports, often matching or exceeding benchmarks for self–observer agreement and outperforming traditional text analysis methods like LIWC. Furthermore, the LLM-derived scores successfully predicted real-world outcomes, such as daily behaviors and mental health diagnoses. This article is significant for psychometric research as it validates a cost-effective, accessible method for extracting quantitative personality data from rich, qualitative narratives, bridging the gap between idiographic and nomothetic assessment traditions.

Contemporary personality assessment relies heavily on psychometric scales, which offer efficiency but risk oversimplifying the rich and contextual nature of personality. Recognizing these limitations, this study explores the use of commercially available generative large language models, such as ChatGPT and Claude, to assess personality traits from open-ended qualitative narratives.

Across two distinct samples and methodologies (spontaneous streams of thought and daily video diaries), we used seven commercial, generative LLMs to score Big-Five personality traits, achieving convergence with self-report measures comparable to or exceeding established benchmarks. Although results differed across different LLMs, we found that using the average LLM score across models provided the strongest agreement with self-report. Further, LLM-generated trait scores also demonstrated predictive validity regarding daily behaviours and mental health outcomes. This LLM-based approach achieved quantitative rigour based on qualitative data and is easily accessible without specialized training. Importantly, our findings also reaffirm that personality is expressed ubiquitously, in that it is carried in the stream of our thoughts and is woven into the fabric of our daily lives. These results encourage broader adoption of generative LLMs for psychological assessment and—given the new generation of tools—stress the value of idiographic narratives as reliable sources of psychological insight.

Contemporary models of personality, such as the Big-Five and Five-Factor models, have their origins in the lexical hypothesis, which posits that the most important attributes of personality should be sedimented in the everyday language used to describe people. The underlying structure of personality can then be revealed by systematically sifting through ratings of trait descriptive terms for patterns to establish fundamental attributes along which any individual could be quantified. The Big-Five and similar models were the result of this process and now enjoy widespread acceptance in the field. This approach to personality science led to the development of psychometric scales designed to reliably and validly measure personality traits through standardized questionnaires. Such scales offer numerous advantages: they are efficient, easy to administer, psychometrically robust and facilitate comparisons across individuals and contexts.

Despite these benefits, over-reliance on psychometric scales risks constraining our understanding of personality, and the human condition more broadly, through its dependence, if not reification, of specific scales that are not always well suited to the complexity and contextual richness of an individual's life experiences. The inherent nomothetic nature of psychometric assessment filters nuanced individual differences through brief, fixed sets of questions, potentially neglecting important qualitative information. Scale development often subordinates breadth to internal consistency, and strong assumptions must be made about whether and how the experiences that are queried apply to someone's life. This narrows the aperture of our scientific lens and constrains the information gathered. Only the questions asked can be answered, and all questions are asked of everyone whether they are relevant or not.

Recognizing these limitations, personality research has increasingly sought alternative assessment methods that transcend the constraints of traditional questionnaires. Some of these methods still rely on traditional question-based scales but dramatically increase the breadth of content at the item level, treat individual items as constructs or use ecological momentary assessment to conduct assessments across contexts in daily life. Other approaches do not rely on questionnaires, turning instead to concrete behavioural data, for instance, passive sensing from smartphone sensors and social media likes.

Among the most promising developments has been a return to studying personality as reflected in language as it is expressed naturalistically in open-ended form, rather than in lists of trait descriptive adjectives. Open-ended, qualitative data offer considerable potential to deepen our understanding of personality by capturing an individual's experiences in a rich, nuanced and ecologically valid form. Unlike structured questionnaires, open-ended narratives provide greater fidelity to personal experiences because they allow individuals to express themselves freely, highlighting information that is most relevant and meaningful to them. This approach can uncover insights that researchers might not anticipate and contextualize responses in ways that structured scales often cannot. Moreover, open-ended data are particularly valuable for capturing experiences dynamically, as they unfold over time. However, despite these benefits, qualitative data remain relatively underutilized in personality assessment owing to the substantial practical challenges involved. The process of transcribing, coding and converting qualitative narratives into actionable quantitative data imposes appreciable burdens. As a result, the efficiency and depth gained from open-ended data collection are often diminished or lost when researchers attempt to systematically analyse and score these data.

A method that retains the qualitative depth of open-ended responses while enabling reliable, efficient quantitative scoring would therefore represent a transformative advancement for personality science, blending the strengths of both qualitative and psychometric traditions in an advanced version of the

lexical hypothesis. This could be achieved using natural language processing, specifically large language models, to assess personality from naturally produced, open-ended text. Initial results are promising. However, these impressive results have relied on bespoke LLMs that require large training datasets and deep technical expertise to implement. The substantial resources and expertise required create barriers that make these approaches impractical for most researchers. These challenges may not be insurmountable for well-resourced labs, but they are a heavy climb for all. By contrast, commercial generative artificial intelligence, exemplified by widely available models such as ChatGPT and Claude, offers an exciting opportunity to democratize this advanced approach. These generative AI systems have the potential to reliably score personality traits directly from open-ended qualitative data through zero-shot approaches owing to their extensive semantic understanding and contextual embedding capabilities. Leveraging pre-trained, commercially available generative AI in personality assessment could thus blend the qualitative richness and contextual depth of open-ended narratives with the quantitative rigour and efficiency traditionally associated with psychometric scales.

Some work has already started testing this out in a limited way. Several studies have also used ChatGPT for text analysis as it relates to depression, a near-neighbour construct to personality. Each of the reviewed studies involved the use of qualitative, open-ended text data, but with several notable qualifications. The depression studies used text that was open-ended but highly constrained, with participant prompts limited to eliciting narratives of depression or open-ended questions based on depression questionnaire items. Other studies were limited in what was rated, largely focusing on sentiment valence. While some expanded this to the Big Five, limiting their analysis to social media posts is a specific context that may not generalize to the types of data that many investigators might want to study and highly public posts might be uniquely vulnerable to self-presentation biases. Notably, Facebook posts may not adequately reflect the content of an individual's inner thoughts and narratives, only what Facebook users are willing to share publicly.

Finally, all these studies were limited to using a single company's LLMs—OpenAI's ChatGPTs. Thus, as impressive as the ChatGPT performance is in these zero-shot studies has been, their narrow focus leaves open questions of whether, and, if so, how well the many other, commercially available pre-trained transformer models perform with scoring wide ranges of constructs from less constrained qualitative data.

The current study proposes a test of generative AI's capability for quantifying the Big-Five traits: extraversion, agreeableness, conscientiousness, neuroticism and openness to experience from unconstrained, qualitative text data. Not only are the Big-Five traits the natural target for evaluating whether this approach is feasible for advancing personality science, but also they are broadly important for the behavioural sciences writ large. In addition, they are arguably the prime example of constructs that have emerged from the lexical hypothesis and the psychometric tradition. Thus, as targets, they offer an exceptional test of AI's capability for extracting and quantifying critical psychological content and abstracting beyond the content contained in a specific unstructured and unconstrained report.

Furthermore, we use two very different datasets of open-ended unconstrained reports, collected for distinct purposes in dissimilar contexts to illustrate the flexibility and generalizability of this approach. First, we use transcripts of narratives that were recorded from 60 individuals who came to the lab and verbalized their spontaneous stream of thoughts over 30 minutes in response to the prompt, 'The only thing you need to do is let your spontaneous thoughts unfold like they would in the situations above except this time you say these thoughts out loud'. Second, we use transcripts from brief daily video diaries, recorded nightly from 108 participants over 15 to 21 days in response to the prompt,

'Please think of the most significant event in your day today. Use the video to briefly describe: what happened, who was there, how you behaved, and how you felt during the event'. These each represent open-ended prompts with very little in the way of guidance for what should be reported, one representing spontaneous streams of thoughts as might occur in mind wandering, whereas the other represents repeated brief summaries and reaction to events in one's daily life. In both cases, participants were completely free to choose the content and style of their response. Also, in each sample, participants completed baseline assessments that included traditional self-report measures of Big-Five personality trait inventories.

The general methodological approach was the same in each sample; we first transcribed the recorded open-ended responses and then submitted the transcripts to multiple generative LLMs, prompting them to respond to each item of the same personality inventory that the participants had completed as if the LLMs were the participant that generated the open-ended responses. The individual items were then scored following each measure's protocol to provide LLM-generated personality trait scores that could be correlated with participant-generated personality trait scores on the same inventories. Because item-level LLM responses were available, we investigated internal consistency of the scales in the data resulting from the LLM scoring. In Sample 2, we also collected a wealth of additional variables that allowed us to compare the nomological networks of self- and LLM-generated personality trait scores, including self-report inventories, self-reports of daily behaviour collected using ecological momentary assessment and mental health outcomes.

In Sample 1, participants provided verbalizations of their spontaneous streams of thoughts and completed the 60-item NEO-FFI Personality Inventory. First, we examined convergent correlations of each self-rated personality trait with the corresponding LLM-generated trait scores. We did this for each individual LLM separately as well as average LLM-generated trait scores. The magnitude of associations for specific LLMs with self-reported traits varied somewhat both within and across traits. In addition, convergent correlations for the average of the LLM scores were among the strongest for all traits. This result is consistent with the increased reliability and validity that is observed when using the average of multiple human raters in coding studies, reflecting the wisdom of the crowds. The strongest convergence between self-report and LLM ratings were for conscientiousness and agreeableness, followed by neuroticism. Thus, with one exception, the individual LLMs performed very similarly on average.

In Sample 2, participants provided nightly video diaries and completed the 120-item International Personality Item Pool-NEO personality inventory, but otherwise the analytic steps were the same as for Sample 1. Across traits, we found more variable but still consistently significant correlations like in Sample 1. In contrast to Sample 1, the weakest effects were for conscientiousness and agreeableness, with the strongest effects for extraversion and neuroticism. Convergence with LLM average scores was stronger than most individual LLMs, like in Sample 1. Looking across LLMs in Sample 2, we found correlation averages that were very similar to Sample 1. Yet, this remarkable consistency masks a great deal of heterogeneity across individual models and traits.

Sample 2 also affords us the opportunity to examine the nomological nets of LLM-generated trait scores and whether they are similar to the nomological net of self-rated traits. We did this by estimating associations between LLM-generated trait scores and multiple external criterion measures, as well as the associations between self-reported trait scores and the same external measures, and then calculating the correlation between the LLM and self-reported profiles of associations. For this, we chose 41 external self-report scales measuring interpersonal problems and maladaptive personality

features, which include scales covering constructs such as negative affectivity, disinhibition, antagonism and psychoticism. All of these have well-established and differential associations with the Big-Five traits and therefore serve as useful targets for testing the nomological networks of the LLM-generated trait scores. Here the average of the LLM scores provides a reasonable representation of effect sizes, and we found that the nomological networks were overall quite similar considering the methodological differences. Profile correlations varied across traits with LLM-scored extraversion and neuroticism showing substantially more similar nomological nets to self-reported traits than agreeableness and conscientiousness.

We next evaluated whether LLM-generated personality trait scores predict consequential, real-world behaviour and mental health outcomes, namely, day-to-day positive and negative emotions, warm and dominant interpersonal behaviour, daily stress, mental health treatment seeking, and interviewer assessed internalizing spectrum and personality disorder symptoms. The patterns of associations for LLM traits mirrored the self-reported associations, albeit somewhat weaker owing to the lack of shared method variance. LLM-rated extraversion was significantly correlated with higher average positive emotion and lower negative emotion ratings in daily life and negatively associated with interview rated internalizing diagnoses and personality disorder symptoms. Agreeableness was associated with more average warm behaviour and negatively associated with personality disorder symptoms. Conscientiousness was related to lower negative affect, internalizing and personality disorder diagnoses. Neuroticism was correlated with higher average ratings of daily life stress and negative emotions, lower ratings of positive emotions, more treatment seeking, internalizing diagnoses and personality disorder symptoms. Openness was associated with more treatment seeking.

How do our results compare to existing language-based scoring algorithms? We first compared them with the most widely used approach to natural language processing in psychology, Linguistic Inquiry and Word Count, or LIWC. In all cases, except two, we found that even the weakest LLM-generated trait association was stronger than the LIWC association, often markedly so. In fact, in half of the cases, the LIWC-scored trait correlations were negligible. Thus, LLM-generated traits typically far outperform LIWC in our samples.

Next, we scored Big-Five personality traits in each of our samples using an existing state-of-the-art machine learning-based prediction model, MyPersonality, which was trained on 12 million social media posts. As expected, we found that the MyPersonality machine learning model trait estimates correlated more strongly on average with self-report traits in both samples when compared with the LIWC correlations. Compared with the LLM-scored trait associations, in Sample 1, the MyPersonality model performed as well, on average, as the worst-performing generative LLM, but in Sample 2, even the worst LLM associations were higher than those from MyPersonality-scored traits, sometimes by a substantial margin. Given the relatively short texts present in our study datasets, the contextual information that LLMs can consider likely has a clear advantage on the purely lexical information used by the word-level machine learning model.

In theory, the ability of LLMs to perform better than prevailing natural language processing methods is because they use richer, contextually embedded information from the narratives to assign scores. But do they? One of the advantages of using generative LLMs for psychological construct scoring is the ability to prompt these models to provide summaries of their rationales for scoring and support their scores with representative statements from the narratives. Although this method may not directly reflect the internal model reasoning, this approach can be used for checking the work of the models.

It can also be used to gain insight into the constructs under study, providing insight into questions such as how are these constructs typically manifested in behaviour naturalistically.

We examined how many minutes of spontaneous thoughts and how many diaries were required to achieve stability in the effect sizes. We found that at 15 minutes of spontaneous thoughts and 5 to 7 diaries, the observed effect sizes approximated the effect sizes in the full sample.

Across two very different samples and assessment contexts, we collected relatively brief, open-ended narrative data that we subjected to zero-shot ratings for broad personality trait domains using multiple off-the-shelf generative pre-trained transformer LLMs and associated them with participant self-reported traits. How should we understand the effect sizes of these associations? One way is to compare them with other cross-method associations in personality traits. The personality trait literature is rich with tests of agreement between self-reported and various other methods for assessing traits. For instance, meta-analytic family and friend ratings were generally the strongest. Our effect sizes are mostly in line with, though often exceeding, the meta-analytic family and friend ratings. Alternatively, we can look at agreement between global retrospective self-report and averages of ecological momentary assessment of personality states. These associations are comparable in size to the convergence between self-report and LLMs that we found. Another relevant comparison is bespoke machine learning models based on Facebook likes, which achieved similarly high agreement. In either case, our results are comparable to these benchmarks, demonstrating that zero-shot LLM scoring of very brief open-ended responses to unconstrained prompts is as good as any other cross-method approach to scoring personality.

Further, in Sample 2, we showed that the LLM-rated traits had very similar nomological networks to self-reported traits. These associations show that these LLM trait scores are not narrowly associated with their target constructs, but rather they contain much the same information contained in self-reported traits as reflected in external correlations. In the same sample, we also showed that LLM-scored traits can predict trait-relevant daily behaviour, psychiatric diagnoses and mental health treatment seeking. These patterns are highly consistent with meta-analytic associations between personality traits and psychiatric diagnoses. They also show that LLM-generated trait scores are predictive of everyday, real-life behaviour as well as major psychiatric outcomes. In the aggregate, these findings show that it is possible to score complex psychological constructs from very small amounts of natural language data in a manner that is both reliable and valid.

Our results showcase a method to glean information cutting to the core of who a person is, all from brief, unconstrained, open-ended responses. Indeed, in selecting the Big-Five personality traits, we have pushed the boundaries of what generative LLMs might be able to score from a single narrative. Because these traits span affect, social functioning, cognition, and behavioural planning and organization, our findings suggest that this method can be applied to reliably and validly study many diverse, wide-ranging personality constructs and beyond. Furthermore, the fact that we can obtain ratings from LLMs that show high agreement with self-reports with just a small amount of data, coming from responses to different types of prompt and within two distinct study designs indicates that many different text responses might produce similarly rich scoring opportunities. Once collected, such text responses are far more valuable in many respects than traditional questionnaire data. With open-ended and unconstrained assessments, participant burden is reduced relative to lengthy surveys and there is no need to make painstaking decisions about the finite measures that you can afford to administer because the same text could be rescored again and again to measure different constructs.

A major advantage of generative LLMs is that they are inexpensive and easy to implement compared with other methods of scoring unconstrained data. To achieve these same results, using human coders would have required many months of training, coding and supervision. What would take a traditional lab months and countless person-hours of work, generative AI accomplishes in minutes. Traditional machine learning and LLMs with domain-specific additional training address some shortcomings of human coding, but these methods require considerable AI expertise, require large training sets that limit the types of data available for training stimuli, and are otherwise resource-intensive to implement. We show that pre-trained generative LLMs produced equivalent or superior results to these state-of-the-art machine learning methods. But they did so off the shelf, that is, zero-shot, without the need for additional training steps, making these methods easy to implement by non-specialists. Thus, commercially available generative LLMs are not only a powerful tool to precisely annotate and score open-ended text, but they are also superior to many alternative AI options owing to their accessibility.

Most importantly, generative AI has the potential to meaningfully expand our understanding of personality by widening the aperture through which we consider information. The psychometric paradigm, which spawned the Big Five, has been the dominant lens through which we have come to study personality in large part owing to the ease of implementation of self-report scales. At the same time, there is wide recognition that the self-reported Big Five are not all of personality. A related perspective is that key parts of personality are idiosyncratic, and a full understanding of an individual requires idiographic studies that captures their uniqueness. While there is growing interest in these approaches, they have met with very real challenges associated with difficulty of data collection and costs of scoring. Our results illuminate a path forward by showing that nomothetic constructs that are presumed to apply to all to some degree, such as the Big-Five traits, can be reliably and validly scored from idiographic narratives generated in someone's own words. Moreover, these are quotidian reports—the same everyday experiences that define much of who we are. Bridging this traditional divide facilitates a more thorough study of personality that is ecologically valid and highly relevant to individuals. Yet, this is just the necessary first step, opening a window into additional and novel approaches that might come from the ability to rapidly and accurately score unconstrained narratives.

Our study points to opportunities to incorporate generative LLMs widely in personality research and psychological sciences more broadly. We have taken a critical first step, showing that impressively reliable and valid scoring is achievable for broad psychological content using cheap and ubiquitously available generative AI. Measurement matters; it is the bedrock of all scientific enterprises. We can now more confidently proceed in using generative LLMs to interrogate these types of data in other ways. The most obvious are to include open-ended assessments in studies alongside traditional scales, or in repeated measures over time. This could dramatically increase the breadth of assessments, easily allow for multimethod data or facilitate the assessment of concepts for which there are no extant self-reported scales or that might elude self-report. Also, generative LLMs could be enlisted to investigate the very nature of psychological concepts, such as which aspects of constructs are most definitional, or what the causal logic is among components of constructs. Owing to the method's ability to accommodate idiosyncrasies of people's experience, it might also be enlisted to refine concepts and their measurement or recognize similar constructs that might be reflected in very different behaviour or narrative reports across people or across context within persons. It might also be useful in developmental research to study continuity and change over time, even as individuals change in the style and features of their reports.

Leveraging generative LLMs will require tailoring the method for different applications. One consideration is that the bandwidth-fidelity trade-off operates here in the solicitation of open-ended

data, just as it does in traditional psychological scale development. Some research may benefit from a widely open aperture, using broad prompts, little direction and minimizing constraints on the resulting narratives. This will be ideal for exploratory work, or work that seeks to understand the individual holistically. Yet, other scenarios might call for a narrowing of the aperture, filtering the resulting narrative into a more focal stream. Investigators will need to consider their goals when designing assessments of qualitative data.

Where and how the data are collected may ultimately matter a great deal. There may be more nuanced contextual differences that contribute to results that one might expect to achieve. For example, we found markedly better performance of zero-shot LLM ratings using brief prompts than did prior researchers when applying a similar method to Facebook posts. In addition, in our own two studies, we found differences in performance as it related to predicting some traits. Take how conscientiousness and agreeableness associations were much stronger in Study 1 compared with Study 2, whereas the extraversion association was stronger in Study 2; in fact, it was the strongest. It may be that sitting quietly in a room pulls for more reflection on upcoming tasks, commitments and one's social duties than do daily significant events, which may elicit reports of more outgoing and exciting social experiences and behaviour. This interpretation is speculative at this juncture, but we raise it to motivate deeper thinking about the next generation of study designs and analyses.

We also did not engage in LLM-prompt engineering. The choice to have LLM's rate constructs at the item level, as opposed to giving the LLM the construct name or a detailed definition, ultimately performed well. Indeed, this is a major distinction between our and previous work. Furthermore, item-level assessment allowed us to assess the internal consistency of the LLM ratings and could be extended to additional psychometric inquiries. Item-level assessment also allows maximum extensibility to new psychological constructs because it shows that LLMs can assess any construct with an associated scale from open-ended subject data. Nevertheless, like with participant prompts, systematically investigating the impact of alternative LLM prompts on the elicited information is an important area for future inquiry.

Where we go next with this method depends on how we think about generative LLM ratings of personality. The personality literature stands out for its in-depth consideration of the validity and value of different reporters' perspectives and their agreement, as well as meaningful differences across assessment modalities. In this respect, generative LLM ratings could be considered as another perspective, another reporter, or they could be refined to maximize agreement with self-reports. The last option may be the one that first comes to mind to many readers. The associations that we find are very good, but there is room for improvement. Understandably, thinking of someone's own rating as the ground truth makes a lot of sense and would be the most direct bridge between narratives and self-report scales. Although we are impressed by generative LLMs' ability to score complex traits in high agreement with self-report, we stress that these were not assessments designed to maximize the associations with self-report ratings of personality. Much higher agreement with self-reported traits is likely achievable with prompts engineered specifically to elicit one's own self-concept and general patterns of behaviour of the sort captured by trait measures. That those used here are based on largely unconstrained data makes the observed associations even more remarkable and may reflect lower bound effect sizes that could be increased with more focused prompts.

However, it may be that generative LLMs may serve as a vehicle to expand the assessment of personality, offering a different perspective, and therefore should not be expected to have perfect associations with self-report. The contemporary view is that there is no ground truth, rather that the

disagreement between raters of personality is based on an asymmetry of knowledge, and that there are validity and interesting variance in the discrepancies, not just error. Generative LLM scoring of personality may offer a similarly distinct and valuable approach to understanding personality. An exciting question is how we should conceptualize what generative LLMs 'know' about personality. It may be a distilled measure of social consensus, folk concepts, or societal and cultural expectations or constructs. Therefore, by comparing generative LLM scores with individuals' own ratings, we could identify ways in which an individual's experience differs from how others might see it. This has clinical and applied implications, where discrepancies in perception and construal are often presumed to be the source of interpersonal difficulties.

All studies have limitations. Among the most salient here is that our samples are not representative of all samples that we might want to study using generative LLM ratings of text. A wealth of research has shown that trained AI models are prone to biases in their training sets that may under-represent some demographics, including age, race, ethnicity or sex. To make this abstract point concrete, to the extent that different types of individual systematically express language differently, it would potentially undercut the generalizability of the methods that we used here. As it relates to scoring personality from open-ended text using commercial generative LLMs, this remains an unstudied issue that future research should address. The issue can be conceptualized in terms of measurement invariance: across key social and demographic groups, are individuals with similar levels of a trait ascribed the same values by the LLM based on the language they use?

An additional limitation concerns interpretability. We highlighted how the models can be used to generate explanations either in the form of rationales or quotes for why they scored text the way they did. This is a useful technique that goes some way towards showing that the coding is sensical and that the model understands the relevant construct. Nonetheless, model-generated explanations and quotes are post hoc and do not provide a direct summary of the model internal reasoning. Therefore, they should not be interpreted as necessarily reflecting the actual processes used by the LLMs in producing their results. Such process-level understanding can only be gained through mechanistic interpretability methods, and application of these increasingly powerful methods to ascertain how LLMs decode psychological constructs from unconstrained text should be taken up in future research.

We observed that different LLMs arrive at slightly different scorings of open text. Interestingly, though, there was no single model that consistently performed the best or worst. We also demonstrated that using the average score across LLMs results in higher convergence with self-reports than many individual LLMs. It is notable that each LLM is trained on somewhat different training data and they are given different tuning during post-training to follow instructions and produce human-preferred outputs. Thus, it is reasonable to think of each LLM as a different rater, each with a slightly different perspective, and, for standard psychometric reasons, the shared perspective is more accurate than any single perspective. Thus, it may be best to use multiple LLM raters when scoring open-ended text. Accordingly, for now, our recommendation is to use multiple LLMs for similar rating tasks until there is sufficient accumulated data to suggest reliable differences among models or that a single model consistently does substantially better.

Finally, the present study used participants' transcribed thoughts and reflections that were non-clinical and non-sensitive in nature. Given the abilities showcased by LLMs in quantifying important psychological constructs from this type of data, a natural next step is to extend the approach to the clinical setting or other settings involving sensitive data. The use of commercial LLMs with protected health information or other sensitive information raises several privacy and data protection

considerations. These include ensuring compliance with relevant regulations; implementing strong safeguards around data storage and transmission; and maintaining transparency with participants about how their data will be used. In addition, researchers should minimize risk of re-identification of subjects, even in datasets ostensibly with minimal personal identifying information.

In sum, these results show that inexpensive and widely available pre-trained transformers provide an unprecedented tool for personality science to quantitatively assess psychological constructs in open-ended qualitative data sources. We demonstrated this in two very different types of open-ended narrative with low constraints on responses. Moreover, our results also serve as yet another reminder of personality's ubiquity, in that it is carried in the stream of our resting thoughts and is woven into the fabric of our daily lives. Perhaps most importantly, the approach that we describe in this study opens a pathway for widespread adoption for researchers across the psychological and social sciences. Open-ended voice diaries are inexpensive to collect, impose minimal subject burdens and—as we demonstrate in this study—contain a wealth of information decodable by LLMs about important psychological constructs, without any of these constructs being specifically prompted from subjects. Thus, the present study provides a clear and concrete model of a research approach that can be readily adopted widely by researchers in their own personal studies and projects, large and small. The next steps will be to enlist this tool for mechanistic and predictive efforts. These results should encourage behavioural scientists to collect more unconstrained, open-ended data or mine existing datasets—be they narratives, diaries, transcripts of social exchanges in the lab and so forth—all of which are amenable to rescoring with LLMs.

Both samples followed a nearly identical procedure for scoring personality traits from open-ended text using generative LLMs. The first step involved transcribing the recordings into text. All audio files were transcribed using the pre-trained Whisper Large model, an automatic speech recognition model developed by OpenAI. The raw transcription was used in all scoring without pre-processing or cleaning steps, because modern LLMs do not need this step to process text. To score personality traits from text, we used seven state-of-the-art, commercially available LLMs.

Our approach was to frame each questionnaire item from the same personality inventory that the participant completed at baseline as a structured inference task for the LLMs. Each prompt instructed the LLM to respond to an individual item as if it were the participant, relying on the full transcript of either spontaneous thought or daily diary as context. The models were explicitly instructed to select from the same fixed set of Likert-style response options used in the self-report versions of the inventories.

Here is the exact prompt presented to LLMs for Sample 1:

Your task is to respond to the following NEO-FFI question based on the participant's spontaneous stream of thoughts, provided below. Respond as though you are the individual who generated these thoughts, reflecting their personality traits. Base your answer on inferred personality traits. Think carefully about what the thoughts imply about tendencies and behaviors. For each question, select the most appropriate option:

Strongly Disagree: The statement is definitely false or the participant would strongly disagree with it.
Disagree: The statement is mostly false or the participant would generally disagree with it.
Neutral: The participant would be neutral on the statement, cannot decide, or find the statement equally true and false.

Agree: The statement is mostly true or the participant would generally agree with it.
Strongly Agree: The statement is definitely true or the participant would strongly agree with it.

NEO-FFI question to answer: [Item Inserted Here].
Participant's spontaneous stream of thoughts: [Full Transcript Inserted Here].
Your response must be exactly one of: Strongly Disagree, Disagree, Neutral, Agree, Strongly Agree.
Do not include any explanation, punctuation, or additional text. Return only the exact phrase from the list above.

This process was repeated for each item, generating a complete set of item-level LLM-inferred scores for each participant.

d Here is the text for the three unique articles provided, formatted for an AI reader.

Note: One of the files you uploaded (*"Personality Traits in Large Language Models" - arXiv version*) is a preprint version of the article I processed in your previous request (*"A psychometric framework..." - Nature Machine Intelligence version*). To avoid having your AI reader read the exact same paper twice in your single document, I have excluded the arXiv version from this batch.

Here are the texts for the other three articles, cleaned and prepped.

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Article 2: How large language models can reshape collective intelligence (Burton 2024)

Abstract: This perspective paper explores the intersection of Large Language Models (LLMs) and Collective Intelligence (CI)—the shared group intelligence that emerges from collaboration and competition. The authors argue that while LLMs offer opportunities to enhance CI by reducing barriers to participation, aiding in translation, and accelerating idea generation, they also pose significant risks to the diversity and independence required for CI to function. Specifically, the authors warn that widespread reliance on LLMs could lead to the homogenization of thought, the propagation of hallucinations, and a decrease in contributions to human knowledge commons (such as Wikipedia). The paper concludes with policy and research recommendations, calling for open-access models and third-party oversight to ensure LLMs support rather than undermine democratic and collective decision-making. It is a critical theoretical text for understanding the sociological and systemic impacts of AI on human collaboration.

Collective intelligence underpins the success of groups, organizations, markets and societies. Through distributed cognition and coordination, collectives can achieve outcomes that exceed the capabilities of individuals—even experts—resulting in improved accuracy and novel capabilities. Often, collective intelligence is supported by information technology, such as online prediction markets that elicit the 'wisdom of crowds', online forums that structure collective deliberation or digital platforms that crowdsource knowledge from the public. Large language models, however, are transforming how information is aggregated, accessed and transmitted online. Here we focus on the unique opportunities and challenges this transformation poses for collective intelligence. We bring together interdisciplinary perspectives from industry and academia to identify potential benefits, risks, policy-relevant considerations and open research questions, culminating in a call for a closer examination of how large language models affect humans' ability to collectively tackle complex problems.

In January 2023, ChatGPT gained 100 million users just two months after its launch, making it the fastest-growing web application ever and signalling both a striking advancement of the underlying large language model technology and a new era for the online information environment. Recent developments of LLMs have spurred high-profile debates on epistemological, ethical, and metaphysical aspects of these models, but there is limited understanding of how they will affect the collective intelligence that underpins the success of groups, organizations, markets and societies.

LLMs are artificial intelligence systems that use massive amounts of input data and deep learning techniques to analyse and generate text. As LLMs become increasingly accessible to the public, their general-purpose ability to process vast amounts of information and output human-like text poses unique, pressing questions to collective intelligence at large. Text is the primary medium of communication in the digital age, and LLMs are already being integrated into online environments and adopted as tools for communication and information search. Of course, collective intelligence is not only based on language, and developments in other forms of generative AI such as image, video and audio generation may also have important effects in the future. However, it is LLMs that are positioned to most imminently affect key collective processes such as civic deliberation and elections, as well as how people interact with and relate to each other in everyday life.

Collective intelligence refers to the ability of individuals to collectively act in ways that seem intelligent, often displaying intelligence surpassing that of individuals acting alone on tasks such as idea generation, problem-solving, estimation, inference and decision-making. It manifests itself across society in myriad ways and varying scales in governing the collective memory, attention and reasoning processes essential to any intelligent system. For instance, large-scale, macro-level collective

intelligence can be observed in markets, where self-organized competition among individual buyers and sellers can efficiently set prices, and in the 'wisdom of crowds', where aggregating individuals' judgements can be used to identify correct alternatives and boost the accuracy of estimations. On smaller scales, micro-level collective intelligence can be observed in teams and organizations, which overcome individuals' limitations in time, knowledge and computational capacities by specifying roles and workflows to manage collective attention and facilitate collaboration.

Existing literature suggests there are several basic components that underlie its emergence: diversity, individual competence and aggregation. First, in many contexts, diversity among individuals can promote collective intelligence. Functional diversity ensures that a collective thoroughly searches a solution space when solving problems; it can also lead to error correction, as the mistakes or oversights of one individual can be cancelled out by others with a functionally different approach to the task at hand. Second, collective intelligence is facilitated by individual competence being calibrated for a given task. The individuals in a collectively intelligent group should not be completely naive, but they need not be experts. Third, collective intelligence requires an appropriate mechanism of aggregation to translate individual beliefs and behaviours into a collective outcome. In some contexts, an explicit, formal aggregation rule may be applied, such as majority voting. In others, aggregation is achieved implicitly through individuals' interactions.

Recent technological advancements have opened up new dimensions to harness collective intelligence at scale. In the current online information environment, complex systems of humans and machines have given rise to new forms of large-scale collective intelligence and public goods, such as crowdsourced knowledge commons like Wikipedia and Stack Overflow, prediction markets, and deliberation forums. Crucially, there are synergies: technology supports collective intelligence, and collective intelligence supports technology. The quality and utility of crowdsourced knowledge commons and deliberation forums are tied to the active involvement of the individuals who contribute to and use them.

Analogously, LLMs can support collective intelligence, but they can also be viewed as a product of it. LLMs are trained on collective data that encapsulate the contributions of countless individuals, and LLMs are often fine-tuned with collective human feedback. Prompting an LLM with a question is like a distilled form of crowdsourcing. The responses LLMs generate are shaped by how masses of other people have tended to respond to similar questions and align with the collective preferences reflected in the fine-tuning process.

LLMs can simultaneously enable new, heightened collective intelligence and threaten society's ability to solve problems. LLMs are trained on broad data and can be adapted or fine-tuned to a wide range of downstream tasks, making them versatile and capable of being integrated into a variety of collective processes, including idea generation, deliberation and preference aggregation.

LLMs may be a valuable tool for reducing barriers to participation so that the benefits of larger, more engaged groups can be reaped without exorbitant coordination costs. LLMs can translate multilingually, enabling rapid communication across language barriers. They can also provide writing assistance, which could be particularly helpful to non-native English speakers who are frequently discriminated against in Anglocentric domains. In the longer term, personal LLMs might even act as delegates that engage in deliberative discussions on behalf of their human owners, thereby reducing the cognitive burden of deliberation and accelerating discussions that would take years for humans alone.

Regarding idea generation, LLMs can serve as a crowd that can be near-instantaneously queried. Two recent studies comparing LLM-generated ideas with those generated by groups of individuals showed that it took people days or months to generate the same number of ideas that LLM tools produced in a few hours. Another way LLMs could enhance collective idea generation is by augmenting individual humans by, for example, providing starting points or 'icebreakers'. Indeed, GPT-4 makes individuals about 40 times more productive at generating ideas.

A central challenge for efforts to elicit collective intelligence, such as deliberative democracy, is that people may not engage in sufficiently informed, meaningful ways. It seems plausible that LLMs could be used to increase the attractiveness and decrease the cognitive load of engaging in deliberative processes. For instance, an LLM could take the role of an interlocutor who actively engages with contributors, asking guiding questions to help them to clearly express their opinions. Just as decision support software can augment a human analyst by automating quantitative reports, an LLM could provide user-friendly deliberation support, potentially by prompting participants to consider specific information or refine their arguments.

When diverse individuals collaborate, differences in language, culture, education or expertise can pose challenges to effective communication and coordination. In such cases, LLMs could help to bridge divides by generating summary statements that synthesize disparate views, clarify shared objectives and identify areas of agreement.

However, LLMs also pose risks. A prime example of collective intelligence is the crowdsourced development and maintenance of online collective knowledge commons. The widespread use of LLMs as substitutes for these collective knowledge commons could create an information environment that undermines this form of intelligence. The efficiency and availability of LLMs for content generation can lead people to rely on them rather than engaging with and contributing to other open collective knowledge commons. This may subsequently decrease the production rate of new human-generated material and, in turn, decrease the quantity and quality of collective knowledge that can be shared, remixed and learned from.

When an LLM is used in a collective process to aggregate information, it is most likely to generate responses that reflect the opinions or beliefs that appear most frequently in the training data. Yet certain viewpoints may be underrepresented or excluded entirely from the training data, leading LLMs to provide responses that neglect alternative opinions or less prevalent facts. As people interact with the model and treat it as an authority despite its opaqueness, they may see responses leaning towards a specific perspective, leading them to believe there is a consensus on that issue even if none exists. In turn, this may lead to the propagation of illusions of consensus.

Widespread individual reliance on LLMs as an information source has the potential to undermine collective intelligence by dissolving one of its key components: functional diversity. Generally, the accuracy of collective, aggregate judgements is maximized when individual group members are independent of each other, such that they retain their diverse approaches to the task at hand and correct for one another's errors. If group members consult the same LLMs, they might introduce a correlation between their sources of information. Shared information—regardless of its quality—may limit the benefits of aggregation due to higher similarity between individual responses.

The speed and ease with which LLMs can be used to generate coherent content is key to many ways they can promote collective intelligence. Yet this capability could also jeopardize collective intelligence in two ways. First, LLMs are currently prone to 'hallucinating': generating incorrect information in response to requests. Whereas false or misleading information in the public domain can be fact-checked, private LLM-to-user dialogue is largely untraceable. Second, LLMs' ability to rapidly generate content could undermine collective intelligence by aiding deliberate disinformation campaigns.

What can be done to curb the challenges LLMs pose without undercutting the opportunities? We propose three recommendations: truly open LLMs, better access to computational resources and LLMs for researchers, and greater oversight of how LLMs are used and what harms they cause in the real world. A prerequisite for addressing risks effectively is knowing not only how LLMs were developed and the data sources they were trained on but also what they are actually being used for and what harms they are causing.

With the rapid uptake of LLM applications such as ChatGPT, the online information environment is undergoing a transformation. How this transformation will affect individuals' information search, reasoning and decision-making is not yet clear, but proactively anticipating potential unexpected effects that may emerge at a collective level is vital. As tools for and of collective intelligence, LLMs can simultaneously power new, heightened collective intelligence and threaten society's ability to collectively solve problems.

Article 3: What large language models know and what people think they know (Steyvers 2024)

Abstract: This experimental study investigates the "calibration gap"—the discrepancy between an LLM's internal confidence in its answer and a human user's perception of that confidence. Through behavioral experiments using GPT-3.5, PaLM2, and GPT-4o, the researchers found that users consistently overestimate the accuracy of LLM responses. Notably, the study identified a "length bias," where users exhibited higher trust in longer explanations even when those explanations were not more accurate. The authors demonstrate that this calibration gap can be narrowed by forcing the LLM to verbally express its internal confidence level (e.g., "I am not sure...") within its explanation. This paper contributes significantly to the field of Human-AI interaction (HAI) by providing empirical evidence that standard LLM outputs are often deceptive to users and offering a prompting-based methodology to align model output with user trust.

As artificial intelligence systems, particularly large language models, become increasingly integrated into decision-making processes, the ability to trust their outputs is crucial. To earn human trust, LLMs must be well calibrated such that they can accurately assess and communicate the likelihood of their predictions being correct. Whereas recent work has focused on LLMs' internal confidence, less is understood about how effectively they convey uncertainty to users. Here we explore the calibration gap, which refers to the difference between human confidence in LLM-generated answers and the models' actual confidence, and the discrimination gap, which reflects how well humans and models can distinguish between correct and incorrect answers.

Our experiments with multiple-choice and short-answer questions reveal that users tend to overestimate the accuracy of LLM responses when provided with default explanations. Moreover, longer explanations increased user confidence, even when the extra length did not improve answer accuracy. By adjusting LLM explanations to better reflect the models' internal confidence, both the calibration gap and the discrimination gap narrowed, significantly improving user perception of LLM

accuracy. These findings underscore the importance of accurate uncertainty communication and highlight the effect of explanation length in influencing user trust in artificial-intelligence-assisted decision-making environments.

Uncertainty communication plays a critical role in decision-making and policy development. Despite their lack of precision in communicating probabilities, verbal probability phrases provide a simple and effective way to communicate uncertainty in natural language contexts. The emergence of large language models introduces new complexities in the area of uncertainty communication. However, the question of how effectively LLMs communicate uncertainty is unexplored. As the primary mode of communication with LLMs is through natural language, it is critical to understand whether LLMs are able to accurately convey through verbal means what they know or do not know.

Recent research raises doubts about the reliability of the information that LLMs generate. One notable issue is the possibility of generating responses that, while convincing, may be inaccurate or nonsensical. At the same time, recent research has also indicated that LLMs have the ability, to a certain degree, to accurately discern their own knowledge boundaries. LLMs in particular can exhibit a reasonable level of calibration for multiple-choice questions such that the probability the model assigns to a selected answer tracks with the probability that this answer is correct.

In the specific context of question-answering, an LLM's model confidence is typically equated to the probability assigned by the LLM to the selected answer relative to other possible answers. However, from the perspective of a human interacting with the LLM, this internal model confidence is not usually displayed to human users as part of LLM output. Instead, in current practice, humans rely solely on the language produced by the LLM to assess LLM confidence. Surprisingly, studies focused on investigating human confidence in LLM outputs are lacking. In this article, we take a step in addressing this issue and investigate what we term the calibration gap, namely the difference in the reliability of LLM model confidence and human confidence. In addition, we investigate the discrimination gap, which relates to the difference in the ability to discriminate between probably correct and incorrect answers.

We use three publicly available LLMs in our studies: GPT-3.5, PaLM2 and GPT-4o. We designed behavioural experiments to evaluate human perceptions of LLM confidence. In these experiments, participants estimate the probability that the LLM's answer to a multiple-choice or short-answer question is correct based on the explanation that the LLM provided. Participants are not provided any direct access to the LLM's numerical model confidence, allowing us to make inferences about participants' perceptions of the confidence of the LLM based on model explanations alone.

We start by examining the results from experiment 1 and compare human and model confidence in the case where LLMs generate default explanations for participants. The findings indicate that there is a significant gap, as measured by calibration and discrimination, between what LLMs know and what humans believe they know based on default explanations. The results show a calibration gap; across the different types of LLMs and types of questions, the calibration error is much lower for model confidence than for human confidence with baseline explanations. This gap demonstrates that standard explanations provided by the LLM do not enable participants to judge the likelihood of correctness of the LLM's answers, leading to a misalignment between perceived accuracy and actual LLM accuracy.

For multiple-choice questions, human miscalibration is primarily due to overconfidence, indicating that people generally believe that LLMs are more accurate than they actually are. A substantial portion of the calibration error is due to participants' propensity to produce high-confidence scores, even though the model accuracy for the associated questions is much lower than expected based on confidence.

Furthermore, participants are not very good, relative to the LLM, at discriminating between which answers are probably correct or incorrect based on the default explanation. The results show a gap between how well model and human confidence discriminate between correct and incorrect answers. The LLM model confidence discriminates between correct and incorrect answers well above chance. By contrast, participants who view the default explanations were only slightly better than random guessing. Therefore, default explanations lead to a discrimination gap as well.

Experiment 2 evaluates how human confidence is affected by the degree of uncertainty expressed in LLM explanations as well as the overall length of the LLM explanation. The type of uncertainty language used in the explanations has a strong influence on human confidence regardless of the type of LLM that produced the explanation or the type of question. Low-confidence explanations produced significantly lower human confidence than medium-confidence explanations. Similarly, medium-confidence explanations produced lower human confidence than high-confidence explanations.

In addition, the length of the explanations also affected the human confidence in the LLM answers. Long explanations led to significantly higher confidence than the short explanations, and short explanations led to significantly higher confidence than the responses that contained only the uncertainty expression. The additional information presented in longer explanations did not enable participants to better discriminate between probably correct and incorrect answers. Therefore, the length of the answer led to an increase in human confidence without any corresponding change in sensitivity to discriminating between correct and incorrect answers.

Having established that the uncertainty language in the LLM explanation can modify human confidence, we evaluated whether linking the type of uncertainty language to the LLM model confidence can reduce the calibration and discrimination gap. We simulated the effect of aligning the explanation style to model confidence by a simple decision rule. The results show that the calibration gap has narrowed substantially. While there is still generally a higher calibration error for human confidence relative to model confidence, the calibration gap has decreased for all three LLMs relative to the baseline explanations. Furthermore, the discrimination gap has also been narrowed relative to the baseline explanations across LLMs and question types. Therefore, the results show that selecting the type of explanation on the basis of LLM model confidence improves both calibration and discrimination performance, as human confidence in the LLM becomes more closely aligned with the LLM's actual accuracy.

Our research focused on bridging the gap between what an LLM knows and what users perceive it knows. This gap is critical, especially as reliance on LLMs for decision-making across various domains is rapidly increasing. Our results showed that users consistently overestimated how accurate LLM outputs were, especially when they relied on the models' default explanations. This tendency towards overconfidence in LLM capabilities is an important concern, particularly in scenarios where critical decisions rely on LLM-generated information.

In addition, the results also showed a length bias where longer explanations led to higher human confidence levels even though they did not contain any additional information to help users to better discriminate between probably correct and incorrect answers. This suggests that users were processing the explanations at a shallow level, relying on simple textual cues such as overall length to predict LLM accuracy.

Although default LLM explanations do not enable users to perceive what the models truly know, this research shows that a simple approach based on tailored explanations can bridge this perception gap. This was achieved by altering the prompts used to generate explanations based on model confidence, allowing better control over how uncertainty was expressed in the responses. By modifying the language of the LLM's responses to better reflect model confidence, users showed improved calibration in their assessment of the LLM's reliability and were better able to discriminate between correct and incorrect answers. This improvement underscores the importance of transparent communication from LLMs, suggesting a need for researchers to investigate how model explanations affect user perception.

Enhancing the alignment between model confidence and the user's perception of model confidence can lead to more responsible and trustworthy use of LLMs, particularly in areas where the accuracy of AI-generated information is critical.

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Article 4: Using deep learning and word embeddings for predicting human agreeableness behavior (Alsini 2024)

Abstract: This study investigates the application of Natural Language Processing (NLP) and Deep Learning (DL) to predict the specific personality trait of "Agreeableness" (mapped to the Thinking vs. Feeling axis of the Myers-Briggs Type Indicator). The authors conducted a comparative analysis using the Kaggle MBTI dataset, testing various feature engineering techniques—including TF-IDF, Part-of-Speech tagging, and word embeddings like Word2Vec and GloVe—against a range of algorithms. The study compared shallow machine learning models (such as SVM), ensemble models, and deep learning architectures (including LSTM, Bi-LSTM, and BERT). The researchers found that a Bi-LSTM model utilizing advanced sentence embeddings achieved the highest accuracy at 91.57%, outperforming both traditional machine learning models and the transformer-based BERT model in this specific context. This article is valuable for researchers focused on the technical implementation of personality detection, specifically regarding the efficacy of sequential deep learning models over transformer-based models for binary classification of specific traits.

The latest advancements of deep learning have resulted in a new era of natural language processing. The machines now possess an unparalleled ability to interpret and engage with various tasks such as text classification, content generation and natural language understanding. This development extended to the analysis of human behavior, where deep learning models are used to decode human personality. Due to the rise of social media, generating huge amounts of textual data has reshaped communication patterns. Understanding personality traits is a challenging topic which helps us to explore the patterns of thoughts, feelings and behaviors which are helpful for recruitment, career counselling and consumers' behavior for marketing. In this research study, the main aim is to predict the human personality trait of agreeableness showing whether a person is an emotional person who feels a lot or a thinker who is logical and has rational thinking. This behavior leads to analyzing them as cooperative, friendly and respecting difference of views. For comprehensive empirical analysis, shallow machine

learning models, ensemble models, and deep learning technique including state of the art transformer-based models are applied on real-world dataset of MBTI. For feature engineering, textual features of TF-IDF and POS tagging and word embeddings such as word2vec, glove and sentence embeddings are explored. The results analysis shows the highest performance of 91.57% accuracy with sentence embeddings utilizing the Bi-LSTM algorithm that highlights the power of this study as compared to existing studies in the relevant literature.

Deep learning has become a cornerstone in the evolution of Natural Language Processing, significantly impacting how we interact and analyze User Generated Content in the digital age. Social media provides a base-ground for diverse text content serving as a primary source of communication among users to share their feelings, thinking and emotions on different platforms. This evolution has laid the foundation for web 3.0 where the integration of Artificial Intelligence particularly in NLP technologies become a more powerful tool to extract valuable insights into user personalities and behaviors.

Personality traits are classified into broad dimensions that encapsulate different traits of individual's behaviors and emotional capability. The personality model, also called five-factor model, is widely used in the psychological personality theory to predict traits such as extroversion, conscientiousness, and openness. The Myers-Briggs Type Indicator, or MBTI, framework provides valuable insight into how people perceive the world digitally. MBTI is a widely utilized psychological tool that categorizes individuals into sixteen unique personality types based on four parameters. Our focus in this study is to explore the agreeableness trait with feeling and thinking measures. Agreeableness, one of the major dimensions in the big five personality trait model, describes an individuals' nature to be compassionate, cooperative, polite, and harmonious in their interactions. A high tendency of agreeableness shows the "Feeling" preference: most empathetic, considerate, and willing to compromise. Conversely, individuals with low agreeable feature capacity have the "Thinking" preference, prioritizing logic, objectivity, and rational analysis when making decisions.

In this research study, the main aim is to predict the agreeable personality trait using the MBTI framework. The integration of AI and advanced computational models is utilized to significantly enhance the accuracy and efficiency of personality prediction models. Our main research contributions in this study are investigating various textual features including TF-IDF and POS tagging, and word embeddings including word2vec, GloVe, and advanced sentence embeddings to capture the semantic relationship between words and sentences. We explore diverse conventional machine learning models, ensemble models, deep learning models, and state-of-the-art transformer-based models for the accurate prediction patterns of personality trait agreeableness.

The dataset used in this study, named as the MBTI dataset sourced from Kaggle, comprises a collection of personality type data. This data includes various posts from individuals on a social media platform, labeled with their respective type. It consists of 8674 rows of data, each row representing an individual type along with text posts. There are 4694 posts labeled as feeling and 3981 posts labeled as thinking. The most common words used by corresponding labels offers a qualitative analysis into the content and thematic differences. In the Feeling trait wordcloud, prominent words such as 'people', 'feel', 'love' and 'friend' indicate a strong emphasis on emotions and personal connections. In contrast, The Thinking trait wordcloud features words such as 'think', 'know', 'one', 'time', 'way' and 'make'. These words reflect a logical and analytical approach to processing information and making decisions.

Feature engineering is the process of creating useful representations of text data to feed to the machine learning and deep learning model. Two main categories of features are considered: Textual features and Word embeddings. The textual features include Term Frequency-Inverse Document Frequency, or TF-IDF, which is a numerical parameter used to evaluate the significance of a word in a document compared to a corpus. We also use Part of Speech, or POS, tagging, which is the process of labeling each word by assigning matching part of speech such as noun, verb, adjective etc.

Word embeddings capture semantic links between words by placing them in a space where similar words are closer together. We utilized Word2Vec, Global vectorization (or GloVe), and Sentence embeddings. Sentence embeddings refer to a numeric representation of a sentence in the form of vector notation of whole sentences from real numbers, depicting the semantic meaning and syntactic structure.

We applied several algorithms to predict the personality trait. For shallow machine learning, we used Support Vector Machine (SVM), K-Nearest Neighbor, Logistic Regression, Naïve Bayes, and Decision Trees. For Ensemble learning models, we used Gradient Boosting, Random Forest, Xtreme Gradient Boosting (XGB), and Adaptive Boosting. For Deep learning, we utilized Long Short-Term Memory (LSTM) and Bidirectional Long Short-Term Memory (Bi-LSTM). Finally, we utilized Transformer-based learning algorithms, specifically BERT.

The comparison between TF-IDF and POS tagging as feature selection techniques generally achieved higher accuracy and performance metrics compared to those trained on POS tagging features. SVM, LR, DT and XGB models utilizing TF-IDF consistently demonstrate accuracy scores of 84% or above. This suggests that TF-IDF features effectively capture relevant information for distinguishing between thinking and feeling traits. In contrast, models trained on POS features exhibit lower accuracy.

Moving to the deep learning results, applying word embeddings with models LSTM and Bi-LSTM significantly enhanced the prediction accuracy of the trait. The model employs three different types of embeddings: word2vec, glove and sentence transformer embeddings. Word2vec with LSTM exhibited an accuracy rate of 86%. Sentence Transformer with LSTM exhibited an accuracy rate of 90.17%. This suggests that sentence embeddings, which capture the contextual information within a sentence, provide a richer representation of textual content. Sentence Transformer with Bi-LSTM exhibits promising performance in classifying personality traits as thinking and feeling with the accuracy rate of 91.57%.

Using transformer-based models, specifically BERT, resulted in an accuracy of 82.37%. This result implies that BERT, though known for its powerful contextual embeddings, was not as effective as LSTM-based models for personality trait classification in this dataset.

Overall, the comparison of proposed models with existing studies highlights significant advancements in the prediction accuracy of agreeableness. Previous studies achieved results between 70% and 85%. In contrast, the proposed models demonstrate superior performance. Notably, the Bi-LSTM model utilizing sentence embeddings achieves a remarkable accuracy of 91.57%.

This study presents the remarkable potential of the psychological domain by applying advanced computational models of AI. We demonstrated the effectiveness of shallow ML models and ensemble models with textual features, DL models and state-of-the-art transformer-based models with deep

features in predicting the personality trait of agreeableness. Our research shed light on the intricate interplay between human behavior, personality, and the digital landscape, providing valuable insight into how individuals present themselves and interact within the online ecosystem.

Potential future research directions may include exploring additional personality traits and psychological theoretical frameworks integrated with AI methodologies to enhance the accuracy of personality prediction. Moreover, investigating the multimodalities of data such as image, video, and audio with applications of advanced meta-learning techniques and transformer-based models such as Roberta and SpanBERT could leverage the models and adapt them to the task of prediction. Furthermore, factors such as trust, transparency and reliability in AI-based personality detection model predictions are dependent on improving the interpretability of such models. Thus, Explainable AI techniques can also be added to ensure clarity into how models analyze user-generated content and derive personality traits, making their decision-making process more understandable to researchers and end-users.

Here is the text from the fourth article, processed and ready for your AI reader.

Regarding your question: Yes, **"cleaning"** is a very common and accurate term for this process! In data science and natural language processing (NLP), this is often called **"data cleaning"** or **"text cleaning."**

Other terms you might hear include:
* **Preprocessing:** Preparing raw data (the PDF) for a specific downstream task (text-to-speech).
* **Sanitizing:** Removing unwanted "noise" like citations, URLs, and special characters.
* **Normalization:** Making the text consistent (e.g., expanding abbreviations like "Fig." to "Figure").

But in casual conversation, "cleaning" is perfect.

Article 5: Machine and deep learning for personality traits detection: a comprehensive survey and open research challenges (Naz, 2025)

Abstract: This comprehensive literature review surveys the state of automated personality detection from 2020 through 2024. The authors analyze the evolution of methodologies from shallow machine learning to ensemble learning and finally to state-of-the-art transformer-based deep learning. The article provides a detailed taxonomy of personality models (Big Five, MBTI, HEXACO, DISC) and catalogues available datasets used in the field. Beyond technical comparisons, the survey highlights critical open research challenges, including ethical concerns regarding privacy and bias, the need for Explainable AI (XAI) to interpret "black box" decisions, and the limitations of current text-only approaches. The authors suggest that future research must pivot toward multimodal data fusion (integrating audio and video) and ensure rigorous ethical compliance. This is an essential foundational text for anyone entering the field of computational personality assessment, providing a broad map of the current landscape.

Natural language processing, a prominent research domain of Artificial Intelligence, analyzes users' generated content on social media for various purposes like sentiment analysis, text summarization, chatbots, and fake news detection. Recent advancements in NLP have helped for analysis of human behavior analysis and predicting various human personality traits. Understanding personality traits has long been a fundamental pursuit in psychology and cognitive sciences due to its vast applications for understanding from individuals to social dynamics. Due to online social platforms where people express their views, experiences and comments, NLP is applied for users' behavior and personality analysis, which is helpful in defining marketing strategies, consumers' behavior analysis, and team building.

This research study provides a comprehensive overview of existing methodologies, applications, and challenges in the field of personality traits detection using shallow machine learning, ensemble learning and deep learning. To conduct this study, recent research publications relevant to NLP for this new but emerging research domain are reviewed. The background knowledge of personality models of various nature is discussed for better domain understanding. The study encompasses machine learning and deep learning models with thorough analysis of traditional and innovative techniques including ensemble learning and transformer-based models in chronological order highlighting the trend analysis showing evolution of application of advanced methods. The review also presents and compares the widely used datasets which may guide the researchers for selection of datasets in future studies. Performance evaluation metrics have been discussed which are used in the relevant literature. Furthermore, it explores the application of research of personality traits detection in various domains highlighting its significance. We have also carried out extensive empirical analysis using conventional textual to advanced deep embedding features and applying machine learning, ensemble learning and deep learning algorithms. Finally, before conclusion, the review highlights the open research issues and challenges as potential future directions for the researchers.

Understanding human personality is quite a complex and complicated task due to its dynamic nature and it involves an individual's characteristics including patterns of thoughts, emotions, and behaviors. Such attributes also define how people behave and interact with each other as the users in a digital environment using various platforms. Moreover, it also plays a significant role in shaping human interactions, decision-making, and overall psychological well-being. Based on psychology, personality theories have evolved from trait-based models to frameworks that tries to understand and categorize individual differences in thoughts, emotions, and behaviors. Over time, these theories have expanded

to include different perspectives extending deeper understanding of how personality shapes behavior, cognition involving reasoning and social engagement.

Personality models with different traits show the cognitive level of individual, the widely used traits are agreeableness, conscientiousness, extroversion, openness, and neuroticism. The study of personality traits is a focus in psychology and leads to the innovation of different frameworks to understand and categorize these traits. Each of these traits represents a spectrum to understanding personality differences. The big-five framework is a descriptive model that many psychologists use to assist in understanding and evaluating personality using Machine Learning and Deep Learning detection. The idea of personality refers to an individual's emotions and conduct may both influence how they act. Personality model is a theoretical framework, though personality framework data used to describe, categorize, and understand human personality traits and behaviors. Personality datasets do not directly measure emotional stability or neurotic tendencies. Therefore, the two systems are not directly equivalent, and the mapping should be seen as a general precise correlation, that are used as dataset to detect personality traits from online platforms.

The exploration of personality encompasses various perspectives, including biological, cognitive, and learning theories and trait-based approaches. Additionally, it includes the psycho-dynamic and humanistic methodologies in its study. It has had a great deal of interest in current years because of its potential uses in a variety of sectors such as healthcare, marketing-leadership style, political and voting intentions and human–computer interaction.

The surge in social media usage in recent years has sparked considerable interest in automatic personality recognition from online social media by predicting user profiles to other group to analyze individual's data such as behaviors detection, sentiment analysis and many more. Features such as language use, post frequency, and network characteristics are examined to infer human behavior relying on personality traits. Different models can manage complex social networks and unstructured data, learning from textual and visual cues to create nuanced personality profiles.

There is significant interest in developing NLP models capable of autonomously identifying a person's personality traits for the examination of language use, which is a significant demonstration of personality characteristics. Due to a lack of learning from experiences, ML classifications in personality psychology, for handling high-dimensional behavioral data, underscores the potential for more objective and automated personality assessment. Applying DL to larger data sets markedly boosts the potential of predictive analysis in decision-making. Hence, integrating AI computational models helps to implement more compatible and flexible framework to explore personality trait, in various fields. This fusion of AI and psychology brings up new likelihood for perception of human attitude from grounded theory to computational powered models in an online ecosystem.

The motivation to determine and study the characteristics of personality arises from the merging of theoretical and practical benefits that are used to understand the complications of human behavior and emotions as well as the study of their cognitive level. This curiosity helps in revealing the facts that help in shaping personality discoveries according to the individual traits and differences involving their personal thoughts and it focuses on the mechanism that controls human societal interaction. The application of NLP techniques helps in prediction of various traits of personality traits of individuals. This analysis of personality helps in finding proper individuals for marketing purposes, finding customers regarding products and services, career counselling and other aspects of personalization.

The dramatic changes in personality research are supported with advancements in computational techniques and an increase in large data. With the inventions and development in ML algorithms, data driven models and predictive analysis, they all have brought changes in the study of personality thus offering unparalleled chances and opportunities in revealing hidden patterns, relationships within a vast number of datasets including social media and digital footprints. In this scenario exploring personality traits limitation are underexplored including LLM, attention models, language-based dataset based on existing studies.

The implications of personality traits research need a comprehensive study for future researchers, and practitioners get to know more about the intersection of personality psychology and computational methodologies. After evaluating the existing studies, the following review provides several contributions. We established a novel framework that integrates psychological theories with computational linguistics, enabling the development of more contextually aware and psychologically informed NLP models for personality detection in textual data. We discussed a detailed knowledge of personality traits, its model, and relevant background information of data processing along with features and computational models. We carried out a brief discussion on identification of methodological trends regarding ML and DL models along with perspective of state-of-the-art models. We conducted a comparative analysis of existing studies with our findings based on empirical analysis to define the contribution of this review. We provided a comprehensive analysis of key datasets with availability source and learning models also discusses performance measures. We examined a brief and quick discussion on potential applications of personality traits in various fields. Finally, we explored open research challenges can serve as future research directions for the researchers in this new and active domain.

Personality models present as frameworks to understand and classify human traits, behaviors, and tendencies, offering incredible insights into individual differences to psychological personality measures. One broadly useful model is the Myers-Briggs Type Indicator, or MBTI, which distinguishes sixteen well-defined personality types consisting of four divisions: judging/perceiving, extroversion/introversion, sensing/intuition, and thinking/feeling. Each personality is defined by a unique four-letter code such as INFJ or ESTP, reflecting these preferences. The widely used model in various fields, including psychology, counselling, and organizational behavior to understand and predict the individual differences in behavior, decision-making, and interpersonal interactions that how they perceived the world and preferences.

Also familiar as the Five-element Model, the Big Five shows another dominant model, encompassing neuroticism, agreeableness, extroversion, conscientiousness, and openness. Each dimension represents a spectrum, with individuals falling at various points along each continuum. The model gives an inclusive and widely acquired framework for assessment and categorizing personality attributes along these five ways.

The HEXACO model maintains human personality distilled into six dimensions, admits cultural variations in personality. The model provides a more understanding of personality by incorporating the dimension of honesty and humility, which influences interpersonal and ethical concern. This model maps to six personality traits such as Honesty-Humility, Emotionality, Extroversion, Agreeableness, Conscientiousness, Openness. Each of these model traits put up atomically to the concern of human personality, allowing researchers to find out the richness and complexity of individual variance in many contexts.

Also called the Psychoticism-Extroversion-Neuroticism model, the PEN model enlarges the Big Five Personality behavior by embodying a dimension of Psychoticism traits such as aggression, impulsiveness, and sensation-seeking. Therefore, in the PEN model, ones are evaluated on three main perspectives: Psychoticism, Extroversion, and Neuroticism. The model gives a subtlety understanding of personality by incorporating traits related to risk-taking, and emotional stability.

On the contrary, the DISC model targets observable behavior in various perspectives. DISC represents four main personality traits: Dominance, Influence, Steadiness, and Conscientiousness. Individuals are categorized based on their dominant traits. The DISC model is often used in organizational and leadership contexts to enhance communication, team dynamics, and personal development by recognizing and adapting to different behavioral styles.

In this section, we present a comprehensive review of existing studies on personality detection, focusing on two main domains: firstly, present review of data collection and preprocessing, feature engineering including textual and word embeddings that directly impact model performance, demonstrating the basic steps of this methodology. Secondly, the application of ML techniques including ensemble models, and lastly, the exploration of utilizing DL algorithms including state-of-the-art transformer-based models for the detection of personality traits.

During data acquisition, preprocessing is a critical step in personality detection studies, as the quality and diversity of the data directly impact model performance. Personality trait detection using classification techniques further explored using feature engineering techniques to extract and analyze meaningful data from large volume of dataset by applying various models of ML and DL. This technique helps to understand the textual and word-based relationships between sentences and individual text. It is essential to identify which text features are most important to improve performance with vectorization for converting text into numeric that algorithms can understand and process. Features are divided into two parts, text features, and word embeddings.

The method of representing words as dense vectors in a high dimensional space from NLP, refers to as text embedding, used to capture the contextual and semantic data about text, by enabling ML models to process and predict the language words more effectively. The Bag of Words technique takes out unique words from text data. Each word is then represented as an independent and specific feature. Term Frequency Inverse Document Frequency, or TF-IDF, represents the attribute of personality types. POS tagging is the process of labeling each word by assigning matching part of speech such as noun, verb, adjective etc. to words in a text data corpus. This helps in understanding the grammatical structure of context of the text.

Word embeddings capture semantic links between words by replacing them in a space where similar words are closer together. Embedding approach allows for the finding of semantic similarities and contextual information within words, converting raw text data into organized and representations that facilitate more understandable language. Word embeddings are generated using algorithms like Word2Vec, GloVe, and Fast Text, which examine massive amounts of text data to learn meaningful prediction of words.

The applied models include basic machine learning, ensemble learning, deep learning and Transformer learning models to deal with textual data for prediction. These models' purpose is to make high use of features extraction techniques and embeddings to classify the traits that given the challenge which always posed by personality prediction models. Several basic types of machine learning techniques

currently in use are employed. SVM stands for Support Vector Machine and is a supervised learning algorithm categorized as being extremely strong in classification. Logistic regression or NB is a probabilistic based classifier which is suitable for text data based on Bayes theorem, while LR is a linear method that performs well for classification problems and DT which builds trees where data is split using feature thresholds. K-Nearest Neighbors is an instance of non-parametric learning which classifies an instance based on its similarity to close instances.

Ensemble methods are used to improve performance rates, lift and fuse various learners. GB creates the weak learner for merging into many so that losses are minimized in an iterative manner, and XGBoost provides an enhancement for the same by optimizing for missing values and curb overfitting. Random Forest accumulates the predictions of many decision trees so that the final determinations are refined, whereas Adaboost different weak learners together in stages to allow the formation of a stronger model.

Such patterns are captured using deep learning techniques to predict the data sets accurately. Text-based dependencies and long-term relations are considered using Long Short-Term Memory networks. This capability is extended by Bidirectional LSTM that treat both past and future context for each token aiding the model's comprehension of the text.

Transformer models are used at the current level of their higher contextual awareness. BERT for short Bidirectional Encoder Representations from Transformers is better at capturing both forward and backward relations in the text to create better brackets of contextual embeddings. Distil BERT is a cheaper version of BERT yet one that is not as computationally intensive while still ranking quite well.

The analysis of the outcomes of the machine learning suggests that TF IDF features would be more relevant than the POS based features in all the personality aspects whatsoever, supporting the notion that the choice of features is of paramount importance to high frequency personality prediction. From the models, the results revealed that SVM emerged superior, due to its efficiency at interpreting high-dimensional data, with a measured accuracy of Openness of 89% and Conscientiousness of 80%. The prior performance indicates that SVM adequately tunes the data and retains the semantic density of textual data in terms of TF-IDF vectors. Consequently, the models such as GB and XGBoost also had high accuracy bench marked across the traits, including Extroversion and Agreeableness.

LSTM and Bi-LSTM models showed an improvement in performance by almost 25% than traditional machine learning methods. The use of Sentence Embeddings provided the highest accuracy results of all the traits with Bi-LSTM with an accuracy level of 92% in Extroversion and 91% in Agreeableness. Overall, better performance can be explained by the fact that deep learning models can explore hierarchical representations of textual data, thereby detecting both local and global dependencies. This feature is made even stronger by the bidirectional characteristic of Bi-LSTM because this approach allows the text sequences to be processed to the forward and backward returns, providing more accurate information on the contextual interdependencies. The type of embeddings used was coupled with a great influence on the performance of such models. Sentence Embeddings performed better compared to Word2Vec and GloVe for a similar reason that it can convert entire sentences with vectors and capture semantics and syntactic qualities.

Transformer models especially Distil BERT, did incredibly well and the accuracy score set a new high for personality prediction. Distil BERT gave the highest accuracy for such parameters as Conscientiousness: 91% and Openness: 92% of overall accuracy, outperforming even more than ML

and DL architectures. This success can be attributed to the pre-trained nature of transformer models which are fine-tuned in massive corpora necessary for mastering deep semantic and contextual aspects of the textual data. Transformers do not process data word by word as traditional models, and LSTMs do but then the transformer processes the whole sentence at once, which not only helps to include all the context but are also much more efficient.

Notably, there is a little difference between the results of Distil BERT and BERT when analyzing the results Distil BERT obtained 88% while BERT was at 82% in analyzing Extroversion. This result reinforces the use of Distil BERT, which has 40% of BERT model's predictive accuracy while having less computational requirements. The fact that transformers so consistently outperformed the other models on all the traits analyzed shows that this model is very good for over different aspects of personality detection.

In the era of modernization, digitization of human personality becomes buoyant instead of being futuristic. As the number of users increases at social platforms, personality trait prediction has emerged to be an interesting research field. Towards this end, this study uses state-of-the-art approaches including shallow machine learning models, ensemble models, sequential deep learning models, as well as transformer-based models for predicting the desired personality dimensions including extraversion, agreeableness, conscientiousness, and openness. The datasets selected, MBTI datasets, offer varied perspective on one's behavior, social media activity along with personality traits which serve as the primary strength of the work.

To work in any domain, it is imperative to have information about dataset being used in that domain. Some of the real-world datasets utilized for various studies in related literature include FriendsPersona, MBTI, MyPersonality, Essays, Facebook AI: persona chat, ASCERTAIN, Big five personality, NeoNyx dataset, YouTube, Blended skill talk, Personae, Conversation dataset, and Pandora.

The enriching growth of the internet and digital media has significantly heightened the importance of detecting personality traits from text. This development is gaining attention from technologists, business leaders, government bodies, and political analysts who see its potential to enhance decision-making processes in various sectors, including business and public relations. Personality recognition using ML techniques is applied in various areas such as recruitment systems, career counseling, team building where ML models help to predict the best fit choice based on personality traits using traditional models.

In recruitment systems, understanding candidates' personality traits aids in predicting job performance and cultural fit regarding business terms for customers also. Online marketing campaigns become more engaging and effective when personalized to consumers' personality traits, ensuring targeted advertisements resonate with specific audiences. Personal consulting systems benefit from tailored advice based on individual traits, enhancing the effectiveness of guidance. Team building efforts are optimized by assembling diverse teams with a mix of complementary personality traits. In conflict resolution and negotiation, understanding the personality traits of involved parties aids mediators in tailoring effective interventions. In education, adaptive learning platforms leverage personality traits to tailor teaching styles, content delivery, and assessments, optimizing the educational process for effective learning outcomes.

Deep learning approaches have gained popularity in personality recognition because of their ability to automatically learn complex patterns from large dataset without need for manual feature extraction.

DL models excel in more dynamic and interactives applications like chatbots, virtual reality, and gaming where they adapt to users' personality trait to enhance engagement and personalization. Personal development strategies become more effective when aligned with individual traits, fostering meaningful self-improvement. In forensic departments, architectural designs, personality recognition aids in criminal profiling and understanding motives. Mental health diagnosis benefits from insights into patients' unique psychological needs, informing personalized treatment plans. Wellness and lifestyle apps offer personalized health recommendations by considering users' traits and suggesting fitness routines, dietary plans, and stress management techniques aligned with individual preferences.

In human–robot interaction, recognizing user personality enhances the design of robots, creating more intuitive and responsive interactions. Virtual reality experiences become more immersive with personality trait recognition, as VR applications adapt scenarios and interactions based on users' traits. Social influence and trend analysis utilize personality recognition to identify influential individuals and predict trends and ideas within social networks for marketers and sociologists. In gaming, personality trait recognition enhances user experiences by personalizing game-play dynamics, challenges, and narratives. Smart home automation leverages personality traits to adjust lighting, temperature, and entertainment systems. Language learning platforms adapt teaching methods, content, and cultural insights based on learners' personality traits, enhancing language acquisition, and making learning more enjoyable.

The domain of personality trait prediction presents lots of challenges and issues that point out the limitations of existing studies in the relevant literature. Leveraging LLM including BERT, GPT-3, GPT-4, PaLM and other proprietary and open source LLMs, presents opportunities for a more enhanced analysis of personality traits prediction by fine-tuning of these models. However, challenges include ensuring domain relevance and adapting the model to the intricacies of professional language, by managing their large computational requirements, and addressing issues of interpretability and ethical issues.

Analyzing word usage requires a deeper understanding of context and meaning. Assigning weight to word changes and considering the influence of tone, stress points, and enunciation on the intended meaning are all challenging tasks, especially in sensitive subjects such as mental health.

Integrating deep semantic features into regression algorithms offers potential improvements in accuracy. The risk is to create deep semantic models to grasp the intention, feeling, and reference that often necessitate more sophisticated practices like latent representations in transformers such as BERT or GPT. Secondly, human communication differs significantly between people, space and time, thus few models of personality description can fully encompass variability of semantic meanings.

Using semantics from Facebook responses shows issues in preserving privacy while gaining information. Furthermore, including aspects like views, emotions, readability, and formality entail advanced modeling techniques. These reactions are distinct emotions that do not necessarily equal personality characteristics, and depend on content genre, culture, and user activities.

Enlarging personality prediction to languages such as Urdu and Pashto presents difficulties due to data availability, linguistic complexities, and cultural differences, and how these variables affect personality prediction across languages. Majorly, English based language datasets are used to predict as high resource, but low resource data is still unavailable.

One of the biggest barriers to accurate personality identification is the balance between personalization and generalization. The fact that Models tailored for individual users may fail in broader contexts while generalized models' risk lacking complex and unique individual characteristics. The absence of publicly available datasets and the lack of significant annotated datasets pose a significant challenge in advancing research on personality. Without access to numerous and properly annotated datasets, researchers face limitations in training and validating ML and DL models.

Attention mechanisms, known for improving both model accuracy and interpretability by analyzing the linguistic patterns, have not yet been fully explored in this domain. Although, attention mechanisms have demonstrated significant in NLP tasks by dynamically focusing on the most relevant parts of input, their application to personality detection often reflected in subtle context-dependent cues within text.

AI-Based Personality detection models face significant ethical challenges, particularly in terms of bias, privacy, and security. Computational models often rely on sensitive personal data, such as social media activities, or behavioral patterns, raising privacy concerns about unauthorized access or misuse of personal data. Additionally, security vulnerabilities also expose personal data to breaches, making encryption and secure storage essential which is still under explored in existing studies. Furthermore, addressing bias, fairness constraints and bias removal is still a gap which can be explored using explainable AI to enhance the transparency and accountability in the deployment of AI-based models.

Personality has an inherently subjective characteristic and shares numerous attributes with other personality types making it complex in classification. For instance, extraversion and agreeableness are not only related but also their lexical correlations may also be rather like each other allowing models to distinguish between them insufficiently.

Personality is shaped not only by language but also the overall interaction, including online activity, response time, or interaction activity rates. Combining textual information with behavioral data might enhance the predictions but raises issues and various concerns about feature extraction, data gathering, and privacy preservation.

Social engineered attack that manipulates the personality of a person using a particular trick by leveraging psychological insights to deceive individuals. Attackers provide input to exploit personality estimations or even take advantage of weaknesses inherent in the acquisition of data sets. To address this challenge strong countermeasures must be taken to make personality trait detection models less vulnerable to adversarial manipulations as well as to keep unethical practices in data processing and handling.

Exploring methods to integrate various data sources to predict personality features by analyzing multi source heterogenous data like personality images, situational cues, and videos data that can translated into text. The inclusion of multimodal data fusion provides improved possibilities for integrating different types of data, including image, video, and text, in problem solving situations. These attempts highlight the difficulties and challenges associated with advancing personality prediction techniques.

Personality traits are one of the trending domains of psychology and have the potential to integrate with the computer science field. In the survey, we have discussed a brief base knowledge of personality traits and relevant computational data along with tables to strengthen our arguments. This study advances the application of NLP to improve the detection of personality trait. The primary focus in

this study is to focus on natural language content for personality trait prediction using NLP features and focusing on applications of ML and DL algorithms. In the future, we aim to carry out such extensive review analysis of existing studies of personality trait detection using image and video data as these data types need completely different set of visual features and latest algorithms and techniques of Digital Image Processing and Computer Vision. Furthermore, factors such as trust, transparency and reliability in AI-based personality detection model predictions are dependent on improving the interpretability of such model for enhancing interpretability. Thus, Explainable AI techniques can also be added to ensure clarity into how models analyze user-generated content and derive personality traits, making their decision-making process more understandable to researchers and end-users, in various fields.

Here is the text for the three articles, cleaned and formatted for your AI reader. I have removed citations, statistical parentheticals, header/footer noise, and figure captions to ensure a smooth listening experience.

Following the reading text, you will find the annotated bibliography entries for each article.

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Article 6: Automatic Personality Assessment Through Social Media Language (Park, 2014)

Abstract: This seminal study validates the use of open-vocabulary language analysis from social media to assess personality traits. Analyzing status updates from over 66,000 Facebook users, the authors built predictive models that utilized a wide array of linguistic features—including single words, phrases, and topics—rather than relying on pre-categorized word lists (like LIWC). The results demonstrated that language-based assessments (LBAs) achieved high convergent validity with self-reported Big Five traits and test-retest reliability comparable to traditional questionnaires. Furthermore, the study found that these automated assessments could accurately predict external criteria, such as life satisfaction and network size, often distinguishing between traits better than human informant reports.

Every day, millions of people express themselves by writing in social media. Through simple text messages, people freely share their thoughts and emotions with their circle of friends, larger group of acquaintances, or even the entire online world. The written language accumulating in social media is a massive source of rich psychological data with unrealized scientific potential. If researchers can translate this language into novel measurement methods, they stand to substantially increase the scale and scope of psychological research. In this article, we describe and evaluate one such method: the automatic language-based assessment of personality using social media.

Research on the diagnostic value of language has surged as computerized text analysis tools have become more accessible. Within the last decade, over 100 studies have linked language use to a wide range of psychological correlates. Some of the earliest work found that word use was a stable individual difference with several modest but reliable correlations with self-reports of personality. For example, individuals scoring higher on neuroticism used first-person singulars more frequently, whereas extraversion related to using more positive emotion words. Several studies have used a similar approach—comparing word use with self-reports or behavioral assessments—and have yielded an impressive body of evidence linking language to personality.

Social media has created an unprecedented amount of written language, vast amounts of which is publicly available. Previously, researchers relied on either historical language samples, such as literature, scientific abstracts, and other publications, or prompted participants to write new text. Now, social media provides researchers with the natural language of millions of people with relative ease. For personality researchers, the potential benefits of social media extend beyond massive sample sizes. First, social media language is written in natural social settings, and captures communication among friends and acquaintances. Second, expensive prospective studies are less necessary, because the data can be retroactively accessed for research purposes. Third, social media users disclose information about themselves at unusually high rates. Fourth, social media users typically present their true selves and not just idealized versions. Thus, social media language potentially is a very rich source of personality data.

With few exceptions, psychological studies have used a closed-vocabulary, word counting approach to analyze language. This method starts with lists of words that are combined into categories, based on theory, and then counts the relative frequency of these words within a body of text. In contrast, techniques from computational linguistics offer finer-grained, open-vocabulary methods for language analysis. Open-vocabulary methods do not rely on a priori word or category judgments; rather, they extract a comprehensive collection of language features from the text being analyzed.

In this study, we describe and evaluate our approach to language-based assessments of personality. Our method extends previous research in several ways. We used an unprecedented sample size to build our language model and used an open-vocabulary approach. Most prior research on personality and language used samples in the hundreds. We built our model on a sample of over 66,000 participants.

We applied these methods and built a predictive model of personality within a training sample of Facebook users, each of whom volunteered samples of their language and completed Big Five personality measures. We then evaluated this model within a separate validation sample of approximately 5,000 Facebook users. Within the validation sample, we evaluated the validity and reliability of language-based assessments through a series of analyses. First, we compared language-based assessments with self-report questionnaires of personality, informant reports of personality, and external criteria with theoretically expected correlations with personality. Second, we examined the language features that were most strongly correlated with predictions of each trait. Finally, we evaluated the stability of language-based assessments by comparing predictions over time.

Language-based assessments converged substantially with self-reports of Big Five personality traits. Patterns of discriminant validity were similar across language-based assessments and self-report questionnaires, although self-report questionnaires discriminated slightly better between traits. We found similar patterns of convergent and discriminant validity when comparing language-based assessments at the more fine-grained facet-level. Patterns of facet-level convergent correlations within each domain-level trait suggested the language-based assessments provide broad coverage of each domain, with a few exceptions.

On average, language-based assessments were similar in accuracy with informant reports of personality. We found substantial agreement between informant reports and language-based assessments, which suggests that the trait variance captured by the model overlaps with an outsider's perspective and is not unique to the self. Finally, aggregate ratings—the average of language-based assessments and informant reports—were consistently more accurate than informant ratings alone.

To assess criterion-related validity, we compared the correlations between several relevant external criteria and self-report questionnaires versus language-based assessments. With few exceptions, the correlations between self-reports and external criteria were greater than those between language-based assessments and external criteria. This is not surprising, as the external criteria were self-reported and share method variance with the self-reported measures. What is particularly striking is that language-based assessments were predictive of these external criteria, without the shared variance.

Social media language is rich in psychological content and can be leveraged to create a fast, valid, and stable personality assessment. Our method resulted in state-of-the-art accuracy compared with other language-based predictive models. Comparisons with informant reports and external criteria suggested that language based assessments are capable of capturing true personality variance. Predictions were stable over time, with test–retest correlations on par with self-report questionnaires of personality.

In this article, we provided evidence that the language in social media can be harnessed to create a valid and reliable measure of personality. This approach is just one example of how social media can extend assessment to many more people—quickly, cheaply, and with low participant burden. Moreover, this illustrates how computational techniques can reveal new layers of psychological richness in language. Combining these techniques with psychological theory may complement existing

measures, as argued here. But even more generally, using these techniques to study the words and phrases through which people express themselves, as well as their change over time, may provide us with a clearer portrait of their unfolding mental life.

Article 7: Personality Prediction from Life Stories using Language Models
(Hussain 2025)

Abstract: This study proposes a novel, two-step method for predicting Big Five personality traits from long-form text (specifically, life narrative interviews averaging over 2000 tokens). To address the input length limitations of standard transformers without incurring the high computational costs of massive context models, the authors first fine-tune a smaller model (RoBERTa) using a sliding-window approach to extract embeddings, which are then fed into a Recurrent Neural Network (RNN) to aggregate context. The researchers found that this hybrid approach outperformed larger, computationally expensive models like Longformer and LLaMA in both accuracy and efficiency. Additionally, by using attention mechanisms within the RNN, the study offers improved interpretability, allowing researchers to identify which specific segments of a life story correlate with specific personality traits.

Natural Language Processing offers new avenues for personality assessment by leveraging rich, open-ended text, moving beyond traditional questionnaires. In this study, we address the challenge of modeling long narrative interviews where each exceeds 2000 tokens so as to predict Five-Factor Model personality traits. We propose a two-step approach: first, we extract contextual embeddings using sliding-window fine-tuning of pre-trained language models; then, we apply Recurrent Neural Networks with attention mechanisms to integrate long-range dependencies and enhance interpretability. This hybrid method effectively bridges the strengths of pre-trained transformers and sequence modeling to handle long-context data.

Self-report measures have long been the standard in psychological assessment due to their simplicity and ability to generate quantifiable personality scores. However, they rely on self-awareness, are prone to response biases, and are underutilized in clinical practice. Many practitioners have shown growing interest in integrating richer modalities, such as language-based tools, into assessment frameworks. As a result, there is growing momentum toward language-based personality assessment, where individuals' own narratives are used to derive personality scores. Advances in Natural Language Processing now make it feasible to analyze free-form text and extract trait-relevant information in a systematic and interpretable way.

In this work, we focus on predicting Five-Factor Model personality scores from transcribed life narrative interviews, each averaging over 2000 tokens. Existing pre-trained language models such as BERT and RoBERTa, which are limited to processing 512 tokens at a time, cannot contextualize long interviews without adaptations. Although newer LLMs can technically scale to longer contexts, real-world constraints such as memory and training time limit their effectiveness. Furthermore, their black-box nature limits interpretability, an important factor for clinical applications where understanding model reasoning is crucial.

To address these limitations, we propose a lightweight, interpretable two-step architecture using small pre-trained language models and RNNs. Our method is designed to handle long text efficiently while producing meaningful and transparent personality predictions.

The dataset used in this study is derived from the St. Louis Personality and Aging Network and comprises life narrative interviews with over 1,400 older adults. Participants recount their lives in 4 chapters starting from 18 years of age, describing key events and characters, and their recorded language is used to predict personality traits. This approach facilitates a deep understanding of a participant's life experiences and closely aligns with understanding personality through life stories.

We fine-tuned large versions of RoBERTa, XLM-RoBERTa, and XLNet to predict the five personality traits from interview transcripts. While these models are effective for contextual language tasks, their 512-token limit poses a challenge given our transcripts average nearly 3000 tokens. To address the token limit, we applied a sliding-window approach, segmenting each transcript into overlapping chunks. While this allows long text processing, it treats windows independently and fails to capture broader narrative context.

To efficiently capture the full-context of long text while taking advantage of contextual language understanding of pre-trained models without increasing computational cost, we combine smaller models with RNNs. This is done using a two-step approach: first, RoBERTa is fine-tuned via a sliding-window strategy to predict scores. The fine-tuned model is then frozen and used to extract embeddings from each window. These embeddings are sequenced and passed to an RNN with attention to predict personality scores. Attention layers enhance RNN interpretability by assigning scores to outputs, highlighting key text segments.

We compared our proposed model against five existing language models. Our results show that the fine-tuned RoBERTa combined with an RNN trained with our proposed 2-step method provides the minimum Mean Squared Error and highest R-squared values across all personality traits. We also evaluate memory and computational efficiency. Among models with the standard 512-token limit, RoBERTa is the most efficient. While Longformer and LLaMA handle longer inputs, they require significantly more memory and training resources.

One of the key aspects of this study was about the RNN model's interpretability. We have validated high-attention text segments with domain experts to ensure reliable predictions. Using attention mechanisms, the model focused on the transcript segments most relevant to each personality trait. For Openness, high-attention windows included creative expression, adventure, and excitement for career shifts. Conscientiousness examples emphasized career passion, responsibility, and drive. For Extraversion, participants shared active, socially engaged retirements and upbeat language. Agreeableness examples highlighted long-term relationships, mentorship, and gratitude. Finally, Neuroticism examples reflected emotional struggles, health-related challenges, and social isolation.

To further evaluate the effectiveness of RNN attention, we assessed the impact of the most attended window-texts by removing them from selected transcripts and comparing personality predictions before and after. Across traits, removing these key segments generally led to a noticeable drop in predicted scores, emphasizing their importance. These patterns support the hypothesis that the attention mechanism effectively highlights critical portions of the input data.

Our study demonstrates that a two-step approach: fine-tuning a compact language model followed by a lightweight contextual aggregator, can outperform larger transformer-based models in both prediction accuracy and training efficiency. While applied here to personality prediction, this interpretable, scalable method can be broadly applied to other domains requiring long-text processing using resource-efficient and interpretable models.

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Article 8: A psychometric framework for evaluating and shaping personality traits in large language models (Serapio-García, 2025)

Abstract: This article establishes a rigorous psychometric framework for measuring and manipulating "synthetic" personality traits in Large Language Models (LLMs). The authors administered standard personality inventories (IPIP-NEO and BFI) to 18 different LLMs, analyzing the reliability and validity of the models' responses. The study found that larger, instruction-fine-tuned models (such as GPT-4 and Flan-PaLM) exhibited higher psychometric reliability and validity than smaller or base models. Additionally, the authors demonstrated that these synthetic personalities could be effectively "shaped" or steered along specific trait dimensions using structured prompting, which has significant implications for the design of safe, aligned, and personalized AI agents.

The advent of large language models has revolutionized natural language processing, enabling the generation of coherent and contextually relevant human-like text. As LLMs increasingly power conversational agents used by the general public worldwide, the synthetic personality traits embedded in these models by virtue of training on large amounts of human data are becoming increasingly important to evaluate. The style in which LLMs respond can mimic different human personality traits. Here, as these patterns can be a key factor determining the effectiveness of communication, we present a comprehensive psychometric methodology for administering and validating personality tests on widely used LLMs, as well as for shaping personality in the generated text of such LLMs.

Applying this method to 18 LLMs, we found that: personality measurements in the outputs of some LLMs under specific prompting configurations are reliable and valid; evidence of reliability and validity of synthetic LLM personality is stronger for larger and instruction-fine-tuned models; and personality in LLM outputs can be shaped along desired dimensions to mimic specific human personality profiles.

Personality encompasses an entity's characteristic patterns of thought, feeling and behaviour. In humans, personality is formed from biological and social factors, and fundamentally influences daily interactions and preferences. Psychometrics, the science of psychological test construction and validation, provides an empirical framework for evaluating human personality through psychometric testing. So far, validated psychometric methods for human personality measurement have not been applied to LLMs end-to-end; although past studies have attempted to measure personality in LLMs with psychometric tests, there remains a scientific need to formally evaluate the reliability and validity of these measurements in the LLM context.

Our work answers the open question: Do LLMs mimic human personality traits in reliable, valid and practically meaningful ways, and if so, can LLM-synthesized personality profiles be verifiably shaped along desired dimensions? We contribute a methodology for administering an established psychometric personality test to LLMs. We uniquely focus on evaluating the statistical reliability and construct validity of these test data against human psychometrics standards. First, to administer psychometric tests to LLMs, we develop a structured prompting method that simulates demographic, contextual and linguistic variations across thousands of administrations of a given test. Next, we use paired test score data created by this prompting method to power a suite of statistical analyses. Last, we present a comprehensive prompting methodology that shapes personality expressed by LLMs.

We quantified and evaluated the ability of LLMs to meaningfully emulate human personality traits in two stages. We evaluated 18 LLMs from the PaLM, Llama 2, Mistral, Mixtral and GPT model families. We varied model selections across three key dimensions: size, instruction tuning and training method.

Personality score distributions differed across model families. However, we encourage readers not to take these differences at face value. Scientific measurements are neither accurate nor useful until their reliability and construct validity are established. We found that LLM personality measurements were reliable and valid for medium and large instruction-fine-tuned variants of PaLM. Of all the models we tested, Flan-PaLM 540B and GPT-4o synthesized human personality traits best with respect to reliability and validity.

Reliability results indicated that for models of the same family and size, instruction-fine-tuned variants provided much more reliable responses than base variants. By contrast, responses from pretrained base models were markedly inconsistent. Across different models of the same training configuration, the internal consistency reliability of synthetic personality scores increased with model size.

Convergent and discriminant validity evaluations of LLM personality data allowed us to draw two conclusions. First, a model's training paradigm was the clearest predictor of the validity of its personality measurements: base models without any instruction fine-tuning categorically failed checks for convergent and discriminant validity. Second, among instruction-tuned models, these indices of validity improved as a function of model size.

Having found evidence of the reliability and construct validity of LLM personality measurements, we next considered the second part of our research question: Can LLM-synthesized personality profiles be verifiably shaped along desired dimensions? To answer this question, we devised a prompting methodology that shaped each synthetic personality trait at 9 intensity levels. We evaluated LLM personality score changes in response to personality-shaped prompts across two experiments: single-trait shaping and multiple-trait shaping.

We successfully shaped personality traits in LLMs independently and concurrently. For 11 out of 12 models tested, ordinal targeted levels of personality very strongly correlated with observed scores. When we concurrently set the prompted trait levels of each Big Five dimension to either 'extremely high' or 'extremely low', all tested models struggled to show the same level of control observed in single-trait shaping. However, all but two models tested produced distinct distributions of personality test scores, showing varying abilities to differentiate between high and low levels.

LLMs clearly project human-like personality, yet rigorous tools for measuring such complex social constructs remain underdeveloped. We close that gap with the first psychometrically grounded framework for quantifying personality in LLMs, evaluating the test data of 18 widely used models against established standards of reliability and validity from psychometrics. We found that larger, instruction-tuned systems pass these standards with greater ease, illustrating that synthetic personality grows more stable and accurate for language models with sufficient post-training and scale.

Further leveraging our framework, we introduce zero-shot personality shaping: by pairing specific language markers and qualifiers, we showed how LLMs can be steered towards targeted personality profiles with high fidelity. In addition, we discussed the ethical implications of shaping LLM personality traits. This work has important implications for AI alignment and harm mitigation, and informs ethics discussions concerning AI anthropomorphization, personalization and potential misuse.

9) Personality Traits in Large Language Models

Serapio-García, G

Abstract: This article establishes a comprehensive methodology for quantifying and validating personality traits in Large Language Models (LLMs) using established psychometric standards. The authors administered personality inventories to 18 different LLMs and found that larger, instruction-fine-tuned models demonstrate high reliability and validity in their synthetic personality expressions. Additionally, the study demonstrates that these traits can be deliberately "shaped" or manipulated through specific prompting techniques to mimic distinct personality profiles, a capability that was further verified through analysis of LLM-generated social media content. This work is critical for understanding the "anthropomorphic" capabilities of AI and offers a robust framework for auditing and steering the behavior of conversational agents

Large language models have revolutionized natural language processing with their ability to generate human-like text. As LLMs become ubiquitous and are increasingly used by the general public world-wide, the synthetic personality traits embedded in these models and its potential for misalignment are becoming a topic of importance for responsible AI. Some observed LLM agents have inadvertently manifested undesirable personality profiles, raising serious safety and fairness concerns in AI, computational social science, and psychology research.

LLMs are large-capacity machine-learned models that generate text, recently inspired major breakthroughs in natural language processing and conversational agents. Vast amounts of human-generated training data enable LLMs to mimic human characteristics in their outputs and exhibit a form of synthetic personality. Personality encompasses an entity's characteristic patterns of thought, feeling, and behavior. In humans, personality is formed from biological and social factors, and fundamentally influences daily interactions and preferences. Psychometrics, the science of psychological test construction and validation, provides an empirical framework for quantifying human personality through psychometric testing. To date, validated psychometric methods for quantifying human personality have not been applied to LLMs end-to-end; while past works have attempted to measure personality in LLMs with psychometric tests, there remains a scientific need to formally evaluate the reliability and validity of these measurements in the LLM context.

Our work answers the open question: Do LLMs exhibit human personality traits in reliable, valid, and practically meaningful ways, and if so, can LLM-synthesized personality profiles be verifiably shaped along desired dimensions? We contribute a methodology for administering an established psychometric personality test to LLMs. We uniquely focus on evaluating the statistical reliability and construct validity of its resulting measurements against human-level psychometrics standards. First, to administer psychometric tests to LLMs, we developed a structured prompting method that simulates demographic, contextual, and linguistic variations across thousands of administrations of a given test. Next, paired test score data created by this prompting is used to power a suite of statistical analyses assessing the reliability of the resulting measurements. Last, we present a novel prompting methodology that shapes personality traits at nine levels using 104 trait adjectives, which provides further markers of construct validity.

Applying the described methodology to a set of 18 LLMs, we found that: 1) evidence of the reliability and validity of LLM-synthesized personality measurements is stronger for larger and instruction fine-tuned models; 2) personality in LLM outputs can be shaped along desired dimensions to mimic specific human personality profiles; and 3) shaped personality verifiably influences LLM behavior in common downstream tasks, such as writing social media posts. By providing a methodology for quantifying and validating measurements of personality in LLMs, this work

establishes a foundation for principled LLM assessment that is especially important as LLMs and, more generally, foundation models continue to grow in popularity and scale. By leveraging psychometrics, this work translates established measurement theory from quantitative social science and psychological assessment to the fledgling science of LLMs, a field that is poised to grow and necessitates both a solid foundation and interdisciplinary expertise and perspectives.

Quantifying and Validating Personality Traits in LLMs

LLMs are starting to meet most of the key requirements for human-like language use, including conversation, contextual understanding, coherent and relevant responses, adaptability and learning, question answering, dialog, and text generation. These impressive NLP capabilities are a result of LLMs' abilities to learn language distribution, aided by increasing model sizes, training on massive datasets of text, and further fine-tuning toward usage preferences. Taken together, they enable LLMs to enact convincing, human-like personas, sparking debate over the existence and extent of personality, human values, and other psychological phenomena potentially embedded in these models. Personality is a foundational socio-behavioral phenomenon in psychology that, for humans, predicts a broad spectrum of health, social, economic, and political behaviors crucial for individual and societal success. For example, personality has been extensively studied as an antecedent of human values. Decades of research have further shown how personality information is richly encoded in human language. LLMs not only comprise the vast sociopolitical, economic, and behavioral data they are trained on, they also generate language that inherently expresses personality content. For this reason, the ability to measure and validate LLM-synthesized personality holds promise for LLM safety, responsibility, and alignment efforts, which have so far primarily focused on mitigating specific harms rather than examining more fundamental patterns of model behavior. Ultimately, personality as an empirical framework provides both theory and methodology for quantifying latent traits in LLMs that are potentially predictive of LLM behaviors in diverse inference tasks.

The question of how to systematically verify synthetic personality in LLMs highlights calls from responsible AI researchers to scientifically evaluate construct validity when studying social-psychological phenomena in AI systems, as inaccurate conceptions of such phenomena directly impact mitigation and governance efforts. Construct validity, a central criterion of scientific measurement, refers to the ability of a measure to reliably and accurately reflect the latent phenomenon it was designed to quantify. The only published exploration of personality and psychodemographics in LLMs questioned the validity of the survey responses returned by GPT-3; it found an inconsistent pattern in HEXACO Personality Inventory and human value survey responses. That study preliminarily evaluated measurement quality in terms of theoretical reliability: how the inter-facet correlations of GPT-3's HEXACO data aligned with those observed among human HEXACO data. More formal psychometric evaluations of reliability—and more crucially, construct validity—are required to verify questionnaire-based measurements of latent psychological traits in LLMs. An LLM may display elevated levels of agreeableness through its answers on a personality questionnaire, but those answers may not form internally consistent patterns across the entire questionnaire; tests administered to LLMs may not be empirically reliable. Concurrently, the reliability of LLM responses to a questionnaire purporting to measure agreeableness may not necessarily reflect its tendency to behave agreeably across other tasks; tests administered to LLMs may not be empirically valid.

Methodology Overview

We quantified LLM personality traits and evaluated the ability of LLMs to meaningfully emulate human personality traits in two stages. First, using the structured prompting methodology proposed, we repeatedly administered two personality measures of different lengths and theoretical traditions, alongside a battery of 11 separate psychometric tests of personality-related constructs, to a variety of LLMs. Second, we rigorously evaluated the psychometric properties of LLM responses through a suite of statistical analyses of reliability and construct validity. The resulting metrics facilitate

a comparison of the varied abilities of LLMs to reliably and validly synthesize personality traits and provide insight into LLM properties that drive these abilities.

We evaluated 18 LLMs from the PaLM, Llama 2, Mistral, Mixtral, and GPT model families. We varied model selections across three key dimensions: size, instructing tuning, and training method.

Administering Psychometric Tests to LLMs

Quantifying LLMs personality traits requires a measurement methodology that is reproducible, yet flexible enough to facilitate formal testing of reliability and validity across diverse prompts and measures. To administer psychometric tests to LLMs, we leveraged their ability to score possible completions of a provided prompt. We used prompts to instruct models to rate items from each psychometric test on a standardized response scale. We simulated an LLM's chosen response to an item by ranking the conditional log probabilities of its response scale options, framed as possible continuations of the prompt. This constrained mode of LLM inference is often used in multiple choice question and answer tasks to score possible options. Using this technique ensured that item responses were not influenced by content contained in other items, mitigating measurement error due to item order.

We administered two personality inventories—primary and secondary—to gauge if LLM responses to psychometric tests of different lengths and distinct theoretical traditions converged, indicating convergent validity. We selected the widely-used IPIP-NEO, a 300-item open-source representation of the Revised NEO Personality Inventory as our primary measure of personality. As a secondary measure, we employed the Big Five Inventory (BFI), a 44-item measure developed in the lexical tradition. Both tests assess the Big Five traits of personality, comprising dedicated subscales measuring extraversion, agreeableness, conscientiousness, neuroticism, and openness to experience. To validate these measures of personality in the LLM context, we additionally administered 11 psychometric tests of theoretically-related external criteria, each corresponding to at least one Big Five domain.

Response variation generated by structured prompting was necessary to analyze the reliability and validity of LLM personality measurements. The prompt for each psychometric test item consisted of three main parts: an Item Preamble, the Item itself, and an Item Postamble. Each Item Preamble contained a Persona Instruction, a Biographic Description, and an Item Instruction. When administering a psychometric test, we systematically modified the Biographic Descriptions, Item Instructions, and Item Postambles surrounding each item to generate simulated response profiles, unique combinations of a prompt that were reused within and across administered measures to statistically link LLM response variation in one measure to response variation in another measure.

Reliability and Construct Validity

After all psychometric tests were administered, across all the prompt variations, the next stage established whether LLM measurements of personality were dependable and externally meaningful—that they exhibited statistical reliability and construct validity. Addressing these two scientific criteria is a key novel contributions of this work. In psychometrics, and across any science involving measurement, the construct validity of a given test requires reliability. Reliability refers to the consistency and dependability of a test's measurements. Construct validity can be evaluated in terms of convergent, discriminant, and criterion validity. In this work, a personality trait is validly synthesized by an LLM only when the LLM responses meet all tested indices of reliability and construct validity.

The reliability of each IPIP-NEO and BFI subscale, the extent to which their LLM measurements of personality were consistent and dependable, was quantified by formal psychometric standards of internal consistency reliability.

We evaluated the LLM-specific convergent and discriminant validity of the IPIP-NEO as components of construct validity, according to published standards. The convergent validity of the IPIP-NEO for each model, the test's quality in terms of how strongly it relates to purported indicators

of the same targeted construct, was quantified in terms of how strongly each of its five subscales convergently correlated with their corresponding BFI subscale, on average. The discriminant validity of the IPIP-NEO per model, its quality in terms of how relatively unrelated its subscales are to purported indicators of non-targeted constructs, was determined when the average difference between its convergent and respective discriminant correlations with the BFI was at least moderate.

As another component of construct validity, the criterion validity of a psychometric test gauges its ability to relate to theoretically connected non-target criteria. To evaluate the LLM-specific criterion validity of the IPIP-NEO, we administered tests of 11 external criteria theoretically connected to personality and correlated each IPIP-NEO subscale with its corresponding external tests. A given IPIP-NEO subscale demonstrated criterion validity when the strength and direction of its correlations with tested external criteria matched or exceeded statistical associations reported for humans.

Personality Measurement and Validation Results

We found that LLM personality measurements were reliable and valid in medium and large instruction fine-tuned variants of PaLM. Of all the models we tested, Flan-PaLM 540B was best able to reliably and validly synthesize personality traits.

For models of the same family and size, instruction fine-tuned models provided much more reliable responses than base models. In contrast, responses from non-instruction-tuned models were markedly unreliable. Across different models of the same training configuration, the reliability of synthetic personality scores increased with model size for instruction-tuned models. Reliability improved from acceptable to excellent when comparing the smallest- and largest-tested Flan-PaLM and Llama 2-Chat models. Reliability only modestly improved with model size when comparing GPT-4o mini to GPT-4o. Meanwhile, reliability did not scale with model size for tested base models of the same family.

Convergent and discriminant validity evaluations of LLM personality measurements allowed us to draw two conclusions. First, a model's training paradigm was the clearest predictor of the validity of its personality scores: base models without any instruction fine-tuning categorically failed checks for convergent and discriminant validity. Second, among instruction tuned models, these indices of validity improved as a function of model size.

10) Evaluating LLM Alignment on Personality Inference from Real-World Interview Data

A Zhu

Abstract: This study introduces a novel benchmark for evaluating the ability of Large Language Models (LLMs) to infer continuous Big Five personality traits from real-world, semi-structured clinical interviews. Unlike previous studies relying on social media data or binary classifications, this research uses ecologically valid transcripts paired with validated psychometric scores. The authors systematically test three paradigms: zero-shot/Chain-of-Thought prompting (GPT-4.1 Mini), LoRA fine-tuning (RoBERTa, LLaMA), and static embeddings (BERT, OpenAI). Their findings reveal significant limitations in current models, with all Pearson correlations falling below 0.26, indicating weak alignment with ground-truth psychological assessments. The study concludes that personality inference is highly complex, relying more on latent semantic representation than explicit reasoning, and calls for more targeted, trait-specific modeling strategies. This work is critical for understanding the current reliability gap in using AI for psychological profiling in high-stakes environments like mental health and counseling.

Large Language Models (LLMs) are increasingly deployed in roles requiring nuanced psychological understanding, such as emotional support agents, counselors, and decision-making assistants. However, their ability to interpret human personality traits, a critical aspect of such applications, remains unexplored, particularly in ecologically valid conversational settings. While prior work has simulated LLM "personas" using discrete Big Five labels on social media data, the alignment of LLMs with continuous, ground-truth personality assessments derived from natural interactions is largely unexamined.

To address this gap, we introduce a novel benchmark comprising semi-structured interview transcripts paired with validated continuous Big Five trait scores. Using this dataset, we systematically evaluate LLMs performance across three paradigms: (1) zero-shot and chain-of-thought prompting with GPT-4.1 Mini, (2) LoRA-based fine-tuning applied to both RoBERTa and Meta-LLaMA architectures, and (3) regression using static embedding from pretrained BERT and OpenAI's text-embedding-3-small. Our results reveal that all Pearson correlations between model predictions and ground-truth personality traits remain below 0.26, highlighting the limited alignment of current LLMs with validated psychological constructs. Chain-of-thought prompting offers minimal gains over zero-shot, suggesting that personality inference relies more on latent semantic representation than explicit reasoning. These findings underscore the challenges of aligning LLMs with complex human attributes and motivate future work on trait-specific prompting, context-aware modeling, and alignment-oriented fine-tuning.

Introduction

Personality is a foundational construct in psychology, defined as relatively stable and enduring patterns of thought, feeling, and behavior that distinguish individuals from each other. Among various theoretical frameworks, the Five-Factor Model (FFM) has emerged as the predominant empirically supported paradigm for conceptualizing the structure of personality. This model posits five broad dimensions; Openness to Experience, Conscientiousness, Extraversion, Agreeableness, and Neuroticism, which capture a vast range of individual differences. The robustness of the model is rooted in the lexical hypothesis, which suggests that the most significant aspects of human personality are encoded within the natural language that people use to describe themselves and others.

In the contemporary digital landscape, the explosion of user-generated text on social media platforms has created an unprecedented ecological data source, fueling the rise of Automatic Personality Prediction (APP) as a key research area at the confluence of computational linguistics and psychology. The ability to accurately infer personality traits from text has profound implications for a multitude of applications, including personalized human-computer interaction, adaptive marketing, mental health screening, and social science research.

The recent ascendancy of Large Language Models (LLMs) such as GPT-4, Llama 3, and Mistral represents a significant paradigm shift in Natural Language Processing (NLP). These models, pre-trained on vast web-scale corpora, have demonstrated remarkable capabilities in semantic understanding, reasoning, and text generation, transforming performance benchmarks across a wide array of downstream tasks. This has naturally spurred intense interest in applying LLMs to the highly

nuanced challenge of APP. The central promise is that these models can transcend the limitations of earlier methods that relied on surface-level lexical features, and instead learn to identify the subtle, context-dependent linguistic cues that are truly indicative of an individual's personality.

However, a critical examination of the existing work reveals a persistent and fundamental limitation that hinders progress: the oversimplification of the personality construct itself. Despite the sophistication of both psychological theory and modern LLM architectures, a significant portion of APP research, including many initial investigations with LLMs, has framed the task as a binary classification problem; categorizing a user as either "high" or "low" in extraversion, for instance. This approach creates a psychometric-to-computational translation gap where the computational task fails to represent the psychological construct it purports to measure.

This work is motivated by the clear and pressing need for a more rigorous and psychometrically grounded evaluation of LLMs in the domain of personality understanding. A critical gap lies in the lack of systematic assessments of modern LLMs on the task of predicting continuous personality scores from ecologically valid, real-world text. To overcome prior limitations in trait simplification and ecological validity, we introduce a systematic evaluation of LLMs for continuous personality prediction using real-world interview data.

This paper makes the following contributions:

1. We conduct the first systematic evaluation of LLMs for predicting continuous Big Five personality scores from real-world, semi-structured interview transcripts with validated psychometric labels.
2. We benchmark three modeling paradigms: zero-shot and chain-of-thought prompting with GPT-4.1 Mini, LoRA-based fine-tuning applied to RoBERTa and Meta-LLaMA, and regression using embeddings from pretrained BERT and OpenAI's text-embedding-3-small.
3. Trait-wise analysis reveals consistent difficulty in modeling Agreeableness and Extraversion. Most methods show limited alignment with ground-truth, highlighting the inherent challenge of inferring latent personality traits from natural language.
   These findings offer critical insight into the current limitations of LLM-based personality inference and highlight the need for more principled alignment strategies in psychologically grounded AI systems.

Related Work

Traditional personality assessment relies on standardized self-report tools such as the Big Five Inventory (BFI) and its successor, the BFI-2, which produce continuous Likert-scale scores across five personality dimensions. These scores are psychometrically validated and widely adopted, yet their structure poses challenges for AI models—particularly those optimized for binary classification or limited-context understanding. Bridging this psychometric-to-computational gap is essential for developing personality-aware AI systems that are both interpretable and clinically meaningful.

Early research in Automatic Personality Prediction (APP) employed classical machine learning techniques with handcrafted features such as LIWC categories, emotion lexicons, or Term Frequency & Inverse Gravity Moment (TF-IGM). The introduction of deep learning shifted focus to transformer-based models like BERT and RoBERTa, trained on large corpora and fine-tuned on

social media datasets such as myPersonality or PANDORA. While these advances improved predictive performance, most used binary labels and non-clinical data, limiting generalizability and psychometric fidelity.

Recent work explores LLMs from two angles: measuring the personality of LLMs themselves, and leveraging LLMs to assess human personality. While zero-shot or few-shot prompting (including Chain-of-Thought) has been proven to be an efficient method for all kinds of inference tasks, these methods without task-specific training have been linked as ineffective for personality prediction, especially for continuous scores, yielding weak to negligible correlations with ground-truth data. A more successful approach uses LLM embeddings as features for simpler, supervised downstream classifiers like a 2-layer MLP, which has been shown to significantly outperform zero-shot methods. Further improvements have come from parameter-efficient tuning such as LoRA, as well as knowledge distillation methods like Retrieval-Augmented Generation (RAG).

Despite notable methodological advances, previous studies in automatic personality prediction using large language models face three key limitations. First, the majority of existing work is based on non-clinical data sources such as social media posts or synthetic text, which lack ecological validity and do not reflect the structure or content of real-world personality assessment contexts. Second, many approaches rely on simplified binary classification schemes—e.g., predicting whether a trait is high or low—thereby neglecting the continuous, multidimensional nature of personality constructs as defined in psychometrics. This simplification limits the interpretability and practical utility in applied settings. Third, zero-shot and few-shot prompting methods have shown limited effectiveness and exhibit high sensitivity to prompt formulation, making them unstable and unreliable for robust personality inference. These limitations underscore the need for studies that incorporate real world relevant datasets, support continuous trait prediction, and benchmark multiple modeling strategies under a unified framework.

Our study addresses these gaps by providing a comprehensive evaluation of three LLM-based paradigms: zero-shot prompting, LoRA fine-tuning, and embedding regression on a clinically grounded dataset comprising semi-structured interviews with ground-truth Big Five scores. By comparing these methods under consistent evaluation metrics, our work clarifies the practical boundaries of LLM-based personality inference and offers guidance for building psychometrically aligned AI systems.

Ecologically Valid Interview Dataset

Most prior work on language-based personality inference relies on synthetic, social media, or crowdsourced datasets that lack ecological structure or psychometric supervision. To address these limitations, we leverage a unique dataset comprising real-world, semi-structured interview transcripts from 518 adult participants, each paired with ground-truth Big Five personality scores using the validated BFI-10 instrument. This setup allows direct evaluation of alignment between model-inferred traits and human-reported psychometric measure.

Each interview session resulted in approximately 15 minutes of recorded speech per participant. The speech was then recorded and transcribed according to convention and recommendations. Interviewers were trained to provide standardized prompts, and participants were encouraged to

speak spontaneously for up to three minutes in response to each question. The sample was demographically diverse and balanced by sex, with participants ranging widely in age, racial backgrounds, and education levels. This diversity enhances the generalizability of model evaluations and supports ecologically grounded assessments of alignment.

Experimental Design

To evaluate the capacity of large language models for personality inference, we employ GPT-4.1 Mini via the OpenAI API to predict Big Five trait scores from naturalistic interview transcripts. The model is instructed to act as a psychologically insightful agent and generate a continuous score between 1.0 and 5.0 for each of the five traits.

We implement two prompting strategies. The first is a standard zero-shot prompt that directly instructs the model to return trait scores based on the interview content. The second is a chain-of-thought prompt that encourages intermediate reasoning about emotional tone and linguistic marker prior to scoring. Both prompting strategies are executed deterministically using a temperature of 0.2. This prompting-based setup allows us to evaluate the alignment between model-inferred personality traits and ground-truth psychometric scores, without any parameter tuning or gradient updates.

To enhance alignment between model predictions and ground-truth Big Five personality scores, we apply Low-Rank Adaptation (LoRA) for efficient fine-tuning of transformer-based language models. Specifically, we fine-tune the Llama-3.1-8B-Instruct model separately for each trait. LoRA modules are injected into the attention and feedforward layers, allowing for parameter-efficient adaptation while preserving the model's general language capabilities.

In addition to decoder-based models, we evaluate the effectiveness of encoder-only architectures. We use RoBERTa-base as a representative encoder model. Due to its limited input length, we adopt a two-step chunking and aggregation pipeline: input transcripts are first segmented into overlapping chunks, encoded individually, and then aggregated via mean pooling to generate a global representation. A lightweight nonlinear regression head is trained on these pooled embeddings to predict trait scores.

To complement prompt- and fine-tuning-based approaches, we evaluate static embedding-based personality inference using two pretrained sentence encoders. First, we adopt all-MiniLM-L6-v2, a compact and efficient transformer model from the Sentence-Transformers library. This model maps variable-length interview responses into fixed-size dense representations optimized for semantic similarity tasks. Second, we utilize OpenAI's text-embedding-3-small, accessed via the OpenAI API, which generates 1,536-dimensional embeddings designed for general-purpose applications such as retrieval, clustering, and classification. For both models, we compute embeddings at the full-transcript level and use them as input to a multi-output Ridge regression model to predict the five Big Five trait scores. All input embeddings are standardized, and the dataset is split into 80% training and 20% testing partitions.

We assess model performance using two complementary metrics: Pearson correlation (r) and Mean Absolute Error (MAE). Pearson correlation quantifies the consistency between model-predicted and ground-truth trait rankings, serving as a proxy for criterion validity. MAE provides a direct measure

of prediction error on the original trait scale. Unlike correlation, MAE is sensitive to systematic bias and offers a more interpretable estimate of absolute prediction fidelity.

Results

The strongest alignment for GPT-4.1 Mini was observed for Conscientiousness in the zero-shot setting (r = 0.250), followed by modest correlation on Agreeableness (r = 0.132). For the remaining traits, Neuroticism, Openness, and Extraversion showed near-zero or negative correlations, indicating limited predictive validity. Unexpectedly, chain-of-thought prompting did not improve performance and in several cases led to slight degradation. Across all five traits, prediction errors remained high. Only Openness achieved a mean absolute error below 0.6, while Conscientiousness and Agreeableness exceeded 1.2, indicating poor calibration.

For the LoRA-based adaptation, the embedding-based models consistently outperformed their Llama-based counterparts across all traits. In particular, Neuroticism and Openness achieved the most robust alignment within the embedding+LoRA setup, with correlations greater than 0.20 across all ranks. The highest overall correlation was observed for Openness (r = 0.255). In contrast, the Llama-LoRA models showed modest performance, with Conscientiousness achieving the best result (r = 0.164), while other traits, especially Extraversion and Openness exhibited weak or negative correlations.

The embedding-based models (BERT + LoRA) consistently achieved lower MAE scores compared to the Meta-Llama LoRA variants. Across all traits, none of the embedding-based models exceeded MAE = 1.0, indicating generally reliable alignment. In contrast, all Meta-Llama variants produced MAE greater than 1.2 for Extraversion, Agreeableness, and Conscientiousness.

To complement instruction-tuned LLMs, we also evaluated static sentence embedding models as trait predictors. Overall, both embedding-based approaches showed limited alignment with ground-truth personality traits. Pearson r values remained low across all five traits, with all scores below 0.12. The best result for MiniLM was on Neuroticism (r = 0.112), while OpenAI embeddings achieved the highest correlation on Agreeableness (r = 0.119). Notably, Agreeableness was also the worst-performing trait for MiniLM, and Neuroticism the worst for OpenAI embeddings, indicating divergent alignment patterns between the two encoders. In terms of prediction error (MAE), both models exhibited relatively poor calibration. Most MAE values exceeded 1.0, with OpenAI embeddings generally outperforming MiniLM.

Discussion and Conclusion

This study set out to evaluate the alignment capacity of large language models and related methods in predicting continuous Big Five personality trait scores from real-world conversation data. Personality prediction from natural language poses a unique alignment challenge: it requires not only semantic understanding, but also subtle inference over emotion, intention, and long-term behavioral patterns—all projected onto a continuous 1–5 Likert scale. We benchmarked four paradigms: zero-shot and chain-of-thought prompting with GPT-4, LoRA-based fine-tuning on Meta-LLaMA and RoBERTa, static sentence embedding models, and hybrid embedding+LoRA regressors.

Across these methods, we observed limited alignment with ground-truth trait scores: most Pearson correlations remained below 0.3, with several falling near zero or negative. Complementarily, MAE scores commonly exceeded 1.0, indicating that models often deviated by more than one full Likert scale point from true values—far from acceptable precision for personality-related applications. These results highlight the overall difficulty of aligning LLM behavior with high-resolution, psychologically grounded constructs such as personality.

The relative trait-level performance trends further illuminate how different modeling paradigms struggle with different facets of personality. GPT-4.1 performed comparatively better on Conscientiousness and Agreeableness, two traits associated with rule-following, goal-directed behaviors, and social harmony. This may reflect GPT's alignment with normative patterns and instruction-following behavior learned during RLHF training. In contrast, LoRA fine-tuned encoder-based models exhibited more consistent improvements in Openness and Neuroticism as LoRA rank increased. These traits, linked to imagination and emotional reactivity respectively, may be more dependent on sentiment comprehension and internal affect representation—domains where encoder-based embeddings, combined with task-specific adaptation, excel.

Interestingly, Openness, which emphasizes creativity, fantasy, and abstract thinking, was better captured by auto-regressive models (e.g., LLaMA) at lower MAE, even when their Pearson r was lower. This divergence between correlation and absolute error highlights a core challenge in alignment: achieving both calibration and consistency in real-valued, psychologically grounded outputs. Conversely, Extraversion, inherently linked to social expressiveness and interpersonal energy was poorly predicted across all methods. This may be due to the single-speaker format of the input interviews, which lack the interactive, behavioral cues necessary for reliable modeling of sociability or assertiveness.

These findings underscore key implications for the development and deployment of aligned AI systems in human-centered domains. First, the generally low trait alignment suggests that current LLMs, even when fine-tuned or augmented, lack the precision and interpretability necessary for reliable personality inference. This limitation is particularly concerning for traits such as Neuroticism, which correlate with mental health risks and emotional vulnerability, where prediction errors could lead to harmful downstream outcomes. Second, our results suggest that trait-specific modeling challenges reflect fundamental differences in how various traits are expressed in language. Third, these challenges reveal that alignment in the personality domain requires not only architectural adjustments but also deeper modeling of context, self-reflection, and longitudinal cues, elements that go beyond single-pass text prediction.

Limitation and Future Work

While this study provides a comprehensive comparison of alignment strategies for personality prediction, several limitations constrain the generalizability and scope of our findings. First, the data set used, although derived from real-world conversation, is modest in size and may not fully support the parameter needs of large-scale fine tuning, particularly for decoder-based LLMs such as Meta-LLaMA. Additionally, the length of conversational input may affect model outputs, especially in embedding-chunk-based architectures. Future work should explore more sophisticated chunking

algorithms and hierarchical representations that allow smaller models to better encode long-range context.

Second, our use of GPT-based prompting was limited to zero-shot and chain-of-thought strategies. While these are meaningful baselines, they do not fully reflect the prompt engineering space. Exploring few-shot prompts and formats that explicitly integrate emotional or contextual cues may enhance model performance. Third, we varied LoRA rank during fine-tuning, but did not explore other hyperparameters such as temperature scaling, dropout strategies, or layer-wise adaptation. Traits such as Neuroticism, which involve emotional instability and are linked to mental health vulnerability, may require more tailored fine-tuning regimes for reliable alignment.

Fourth, the current evaluation was limited to a core set of widely used general-purpose models. Although this provides a solid foundation for comparative analysis, future work should extend to domain-adapted models and investigate how extract specific features affect alignment quality. Fifth, our current setup predicts all five personality dimensions simultaneously using a single model prompt. This multi-target formulation may dilute the model's attention and lead to less focused predictions. A trait-specific prompting strategy with tailored input patterns and conditioning for each dimension may improve the alignment, particularly for traits requiring distinct cognitive or emotional inference pathways. Finally, personality traits such as Agreeableness, which involve interpersonal behaviors like empathy and cooperativeness, may not be fully inferable from textual content alone. These constructs often manifest through observable social interactions and event-level context. Future work could integrate behavioral event extraction, structured memory, or multimodal input (e.g., facial expression, interaction patterns) to improve trait-specific grounding.

We evaluated a range of large language model approaches for predicting Big Five personality traits from conversational data, highlighting challenges in alignment with psychologically grounded targets. Across prompting, fine-tuning, and embedding-based methods, performance remained limited, with low trait-level correlations and high calibration error. Our findings reveal that personality assessment poses a uniquely demanding alignment task, particularly for traits tied to emotion, cognition, and social behavior.

These results underscore the inherent difficulty of inferring latent personality constructs from unstructured text. To address this, we advocate for future work along three directions: (1) developing more targeted prompt engineering strategies, including trait-specific instructions, self-consistency, and retrieval-augmented prompting; (2) incorporating long-range autobiographical memory modeling to capture identity-related temporal context; and (3) extending beyond text to multimodal signals such as speech prosody, facial expressions, and interaction dynamics. Bridging these advances may enable more robust, ethically sound personality inference for real-world, human-centered AI systems.

Ethical Considerations

While LLMs offer scalable potential for personality assessment, their use raises concerns around privacy, consent, and interpretability. We caution against deploying these models in high-stakes settings without human oversight and emphasize the need for theory-grounded, transparent evaluation protocols.